

General Instructions for QP Framing

1. **Mention the Regulations pertaining to the course clearly viz., R2019 or R2024 or R2019/ R2024 (for courses common to both R2019 and R2024)**
2. **Kindly frame the Questions which assess the learning outcomes with utmost care without grammar/spelling errors. Ensure all the necessary data/Figures/Tables etc. are included clearly in every question.**
3. **For each Question, mention CO1, CO2 etc. for Course Outcomes. An Either or type Question should have same CO in both (a) and (b) parts.**
4. **For each Question, mention the Difficulty Level (any no. from 1 to 5) with 1 as Most Easy and 5 as Most Difficult.**
5. **DO NOT Copy and Paste Equations as Images. Type with appropriate tools.**
6. **Marks Split up must be provided for subdivisions (if any) in Part B & Part C.**
7. **Type the Answer Key in the same QP template across the respective Q.Nos.**
8. **Kindly mention the Knowledge Level for each Question as per the following table:**

Knowledge Level (Blooms Taxonomy)					
K1	Remembering (Knowledge)	K2	Understanding (Comprehension)	K3	Applying (Application of Knowledge)
K4	Analysing (Analysis)	K5	Evaluating (Evaluation)	K6	Creating (Synthesis)

9. **For PG Courses, change the Exam Name as M.E. or M.B.A. End Semester Examinations.**

SET 1**QP Setter Name / Dept:**

QP Code:

Reg. No

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SAVEETHA ENGINEERING COLLEGE

(Autonomous, Affiliated to Anna University, Chennai)

B.E./B.Tech. End Semester Examinations – Apr/May 2026 (R2019/R2024)

Common to AI/ML, AI/DS, CSE, IT (Branches)

19AI405/CS3405-FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE

Time: Three hours

Maximum marks: 100

Answer All Questions

PART A**(10 x 2 = 20 marks)**

		CO		
1.	What is the role of sensors and actuators in an intelligent agent? - Sensors capture external inputs (such as light, sound, temperature, or data), forming the basis for the agent's understanding and reasoning about its environment. - Actuators translate the agent's chosen actions into physical or virtual changes, allowing it to achieve its goals within the environment.	CO1		
2.	What is a problem-solving agent? - A Problem-Solving Agent is an intelligent agent that decides what actions to take by searching for a sequence of actions that leads from the initial state to a goal state. - According to Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig: - A problem-solving agent formulates a goal and a problem, then searches through possible states to find a solution. Example: - Route finding (like finding shortest path from Chennai to Coimbatore).	CO1		
3.	Define partially observable environment. - A Partially Observable Environment is an environment in which the agent cannot perceive the complete state of the environment at a given time due to limited or noisy sensors. - In such environments, the agent must make decisions based on incomplete information. - According to Artificial Intelligence: A Modern Approach, an environment is partially observable if the agent's sensors do not give access to the complete state. Example: - Poker game (players cannot see opponents' cards) - Self-driving car in foggy weather Agent maintains a belief state (internal memory of possible states) to handle uncertainty.	CO2		
4.	Differentiate between exploration and exploitation in local search. <table><tr><td>exploration</td><td>exploitation</td></tr></table>	exploration	exploitation	CO2
exploration	exploitation			

	<table><tr><td>Searching new and different areas of the search space.</td><td>Focusing on the best solution found so far.</td></tr><tr><td>Helps avoid getting stuck in local maxima/minima</td><td>Improves the current promising solution.</td></tr><tr><td>Encourages diversity in search.</td><td>Intensifies search around a good state.</td></tr><tr><td>Example: Random restart in hill climbing.</td><td>Example: Moving to the best neighboring state.</td></tr></table>	Searching new and different areas of the search space.	Focusing on the best solution found so far.	Helps avoid getting stuck in local maxima/minima	Improves the current promising solution.	Encourages diversity in search.	Intensifies search around a good state.	Example: Random restart in hill climbing.	Example: Moving to the best neighboring state.	
Searching new and different areas of the search space.	Focusing on the best solution found so far.									
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Encourages diversity in search.	Intensifies search around a good state.									
Example: Random restart in hill climbing.	Example: Moving to the best neighboring state.									
5.	Define Game Theory and mention its applications with an example. Game Theory is the study of strategic decision-making in situations where multiple players interact and each player’s outcome depends on the actions of others. It is used to determine optimal strategies in competitive or cooperative environments. According to Artificial Intelligence: A Modern Approach, game theory analyzes multi-agent environments where agents choose actions to maximize their utility. Applications: Game playing in AI (Chess, Tic-Tac-Toe) Economic decision making Auctions and bidding systems Cybersecurity strategy planning Example: In Chess, each player chooses moves to maximize their chance of winning while considering the opponent’s moves.	CO3								
6.	How does the Monte-Carlo Simulation estimate the value of PI. The Monte-Carlo Simulation estimates the value of π by randomly generating points inside a square and counting how many fall within an inscribed circle, then using the ratio between the number inside the circle and total points to approximate π.	CO3								
7.	What is propositional theorem proving? Propositional Theorem Proving is the process of determining whether a conclusion logically follows from a given set of propositional logic statements (premises). It checks whether a sentence is logically entailed by a knowledge base using formal proof methods. It involves applying inference rules (such as resolution) to derive conclusions from propositional sentences. Methods Used: • Truth table method • Resolution • Forward chaining • Backward chaining Example: If 1. $P \rightarrow Q$ 2. P Then we can prove Q using Modus Ponens.	CO4								
8.	For the sentence "Some philosophers were mathematicians.", write a well-formed formula in predicate logic. $\exists x[Philosopher(x) \wedge Mathematician(x)]$ (There exists at least one xx such that xx is a philosopher and a mathematician).	CO4								
9.	Define knowledge representation.	CO5								

	Knowledge Representation (KR) is the method of encoding information about the world in a structured form so that an AI system can use it to reason, infer, and make decisions. It enables an agent to store facts, rules, and relationships in a way that supports logical reasoning. It involves defining formal languages and structures to describe the world and support intelligent behavior. Common Forms of Knowledge Representation: • Propositional Logic • First-Order Logic • Semantic Networks • Frames • Ontologies Purpose: To help AI systems think, reason, and solve problems like humans.	
10.	State the significance of Partial Order Planning algorithms in classical planning. Partial Order Planning algorithms create plans where the order of actions is not totally specified, allowing for flexibility, concurrency, and more efficient solutions by resolving only necessary ordering constraints between actions in classical planning	CO5

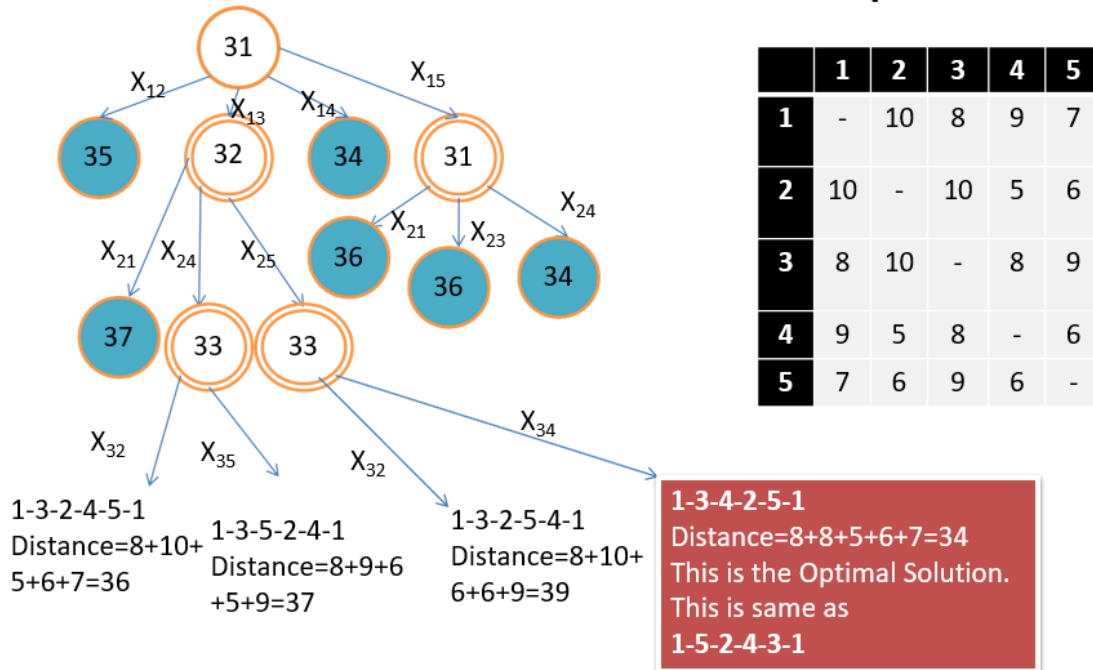
PART B

(5 x 13 = 65 marks)

			CO
11.	(a)	Explain the concept of Artificial Intelligence and Intelligent Agents. Discuss the types and key characteristics of intelligent agents. Concept of Artificial Intelligence – 3 Marks • Definition • Goals • Applications (any 2) Concept of Intelligent Agents – 3 Marks • Definition of agent • Agent–Environment interaction • Rational agent concept Types of Intelligent Agents – 5 Marks • Simple Reflex Agent • Model-Based Reflex Agent • Goal-Based Agent • Utility-Based Agent • Learning Agent (Explain each with 2–3 lines and example) Key Characteristics of Intelligent Agents – 2 Marks • Autonomy • Rationality • Reactivity • Proactiveness	CO1
		(OR)	

	(b)	Explain the characteristics and types of AI task environments with suitable examples. Introduction to Task Environment – 2 Marks • Definition of task environment • Mention PEAS concept Characteristics (Properties) of Task Environment – 8 Marks Explain any 5–6 properties with examples: • Fully vs Partially Observable – 1½ Marks • Deterministic vs Stochastic – 1½ Marks • Episodic vs Sequential – 1½ Marks • Static vs Dynamic – 1½ Marks • Discrete vs Continuous – 1 Mark • Single-Agent vs Multi-Agent – 1 Mark (Each property must include definition + example) Types of AI Task Environments – 3 Marks • Single-agent environment • Multi-agent environment • Competitive / Cooperative environment (Brief explanation with example)	CO1																																				
12.	(a)	Solve the Travelling Salesman Problem (TSP) for salesman visiting 5 cities using the Branch and Bound method. <table><tr><th></th><th>1</th><th>2</th><th>3</th><th>4</th><th>5</th></tr><tr><th>1</th><td>-</td><td>10</td><td>8</td><td>9</td><td>7</td></tr><tr><th>2</th><td>10</td><td>-</td><td>10</td><td>5</td><td>6</td></tr><tr><th>3</th><td>8</td><td>10</td><td>-</td><td>8</td><td>9</td></tr><tr><th>4</th><td>9</td><td>5</td><td>8</td><td>-</td><td>6</td></tr><tr><th>5</th><td>7</td><td>6</td><td>9</td><td>6</td><td>-</td></tr></table> Given the travel cost matrix, determine the least-cost Hamiltonian circuit.		1	2	3	4	5	1	-	10	8	9	7	2	10	-	10	5	6	3	8	10	-	8	9	4	9	5	8	-	6	5	7	6	9	6	-	CO2
	1	2	3	4	5																																		
1	-	10	8	9	7																																		
2	10	-	10	5	6																																		
3	8	10	-	8	9																																		
4	9	5	8	-	6																																		
5	7	6	9	6	-																																		

Branch and Bound-Steps



(OR)

- (b) i) Develop a Python program for Monte Carlo Tree Search (MCTS) in a non-deterministic environment. (7 Marks)
- ii) Write a pseudo-code for an online search agent that learns the environment while navigating using heuristic functions. (6 Marks)
- Monte Carlo Tree Search (MCTS) is a heuristic search algorithm used for decision-making in games and other non-deterministic environments. It balances exploration (trying new moves) and exploitation (using known good moves).
- Below is a Python implementation of MCTS for a simple non-deterministic game where a player moves on a 1D board towards a goal. Random factors affect the outcomes of moves.
- ```
import math
import random

class Node:
 def __init__(self, state, parent=None):
 self.state = state
 self.parent = parent
 self.children = []
 self.visits = 0
 self.value = 0

 def is_fully_expanded(self):
 return len(self.children) > 0

 def best_child(self, exploration_weight=1.41):
 """Selects the child node with the highest UCT (Upper Confidence Bound) value."""
 return max(self.children, key=lambda child: (child.value / (child.visits + 1e-6)) + exploration_weight * math.sqrt(math.log(self.visits + 1) / (child.visits + 1e-6)))

 def expand(self):
 """Expands a new child node with a random state transition."""
```

CO2

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <pre> new_state = self.state + random.choice([-1, 1, 2]) # Random move child_node = Node(new_state, parent=self) self.children.append(child_node) return child_node  def backpropagate(self, result):     """Propagates the result back up the tree."""     self.visits += 1     self.value += result     if self.parent:         self.parent.backpropagate(result)  def simulate(state):     """Simulates a random rollout from the given state."""     goal = 10     while state != goal:         state += random.choice([-1, 1, 2]) # Non-deterministic moves         if state &gt;= goal:             return 1 # Win         elif state &lt;= 0:             return 0 # Loss     return 1 # Goal reached  def monte_carlo_tree_search(root, iterations=100):     """Runs MCTS for a fixed number of iterations."""     for _ in range(iterations):         node = root         # Selection: Traverse the tree using UCT         while node.is_fully_expanded():             node = node.best_child()          # Expansion: Add a new child node         node = node.expand()          # Simulation: Perform a rollout         result = simulate(node.state)          # Backpropagation: Update the tree         node.backpropagate(result)      return root.best_child(exploration_weight=0) # Return best move </pre> <p><b>Explanation</b></p> <p>Selection – Moves down the tree using the Upper Confidence Bound formula.</p> <p>Expansion – Creates new nodes to explore new game states.</p> <p>Simulation – Simulates random plays from the new state.</p> <p>Backpropagation – Updates the node values based on outcomes.</p> <p>Handles Non-Determinism – The random factor in simulate() allows handling of uncertain environments.</p> <p>ii) Write a pseudo-code for an online search agent that learns the environment while navigating using heuristic functions. (6 Marks)</p> <p>An online search agent learns about the environment while navigating. It must explore unknown paths and use heuristic knowledge to make intelligent decisions.</p> <p>Pseudo-code for an Online Search Agent Using Heuristics</p> |  |
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Function ONLINE-SEARCH-AGENT(Start, Goal, h)

Initialize:

CurrentState  $\leftarrow$  Start

Explored  $\leftarrow \{\}$  // Set of visited states

While CurrentState  $\neq$  Goal do:

Add CurrentState to Explored

PossibleActions  $\leftarrow$  ACTIONS(CurrentState) // Get available moves

BestAction  $\leftarrow$  NULL

BestHeuristic  $\leftarrow \infty$

For each Action in PossibleActions do:

NewState  $\leftarrow$  TRANSITION(CurrentState, Action)

If NewState not in Explored then:

HeuristicValue  $\leftarrow$  h(NewState)

If HeuristicValue  $<$  BestHeuristic then:

BestAction  $\leftarrow$  Action

BestHeuristic  $\leftarrow$  HeuristicValue

If BestAction = NULL then:

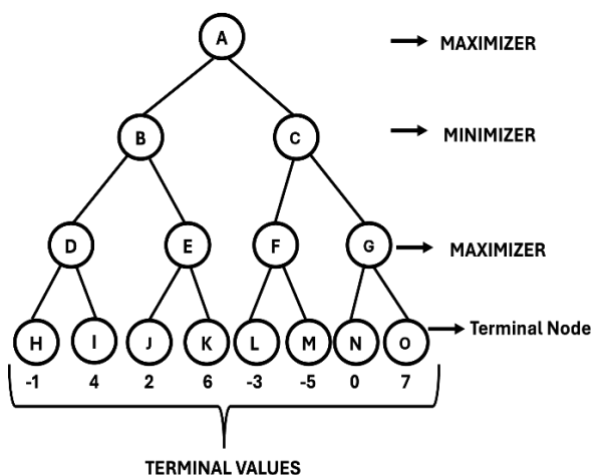
Return "No Path Found" // Dead-end reached

CurrentState  $\leftarrow$  TRANSITION(CurrentState, BestAction)

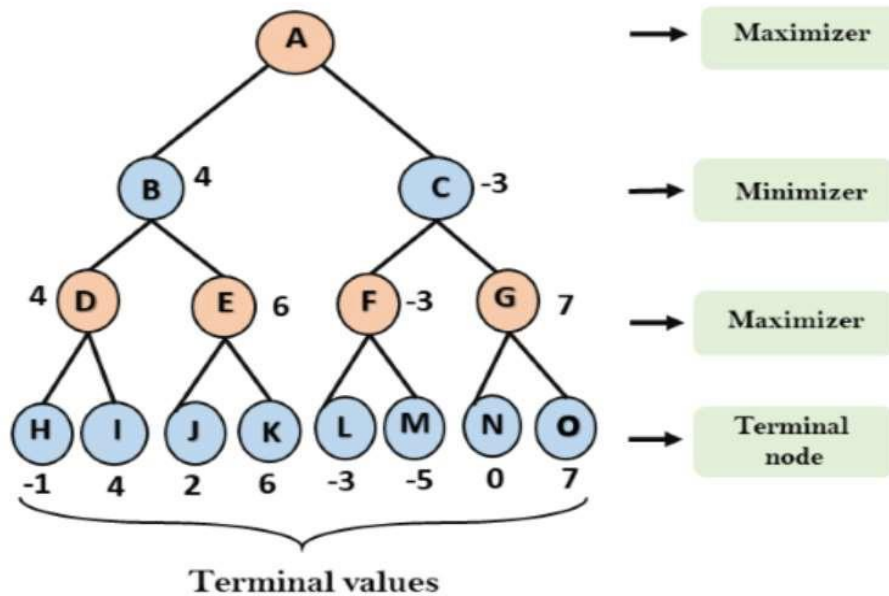
Return "Goal Reached"

13. (a) Perform the Minimax Search Strategy for the following graph:

CO3







- For node A  $\max(4, -3) = 4$

(OR)

- (b) i) Formulate and solve the given cryptarithmic problems as constraint satisfaction problems.  
 CROSS + ROADS = DANGER  
 EAT + THAT = APPLE  
 BASE + BALL = GAMES  
 LET + LEE = ALL  
 (7 Marks)
- ii) Discuss the backtracking and constraint propagation methods used for solving CSP-based cryptarithmic problems.  
 (6 Marks)

**ANSWER:**

- Problem formulation as CSP (Variables, Domains, Constraints) – **3 Marks**
- Showing constraints (all-different + carry constraints) – **2 Marks**
- Correct final solution (any two fully solved clearly) – **2 Marks**

**General CSP Formulation (Applicable to All)**

A cryptarithmic problem is modeled as a **Constraint Satisfaction Problem (CSP)**:

- **Variables:** Each distinct letter
- **Domain:**  $\{0-9\}$
- **Constraints:**
  - All letters must have different digits (All-Different constraint)
  - Leading digit  $\neq 0$
  - Column-wise addition constraints with carries

**CROSS + ROADS = DANGER**

One valid solution:

C = 9  
 R = 6  
 O = 2  
 S = 3  
 A = 5  
 D = 1  
 N = 8

CO3

|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
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|     |     | $G = 7$<br>$E = 4$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
|     |     | <b>EAT + THAT = APPLE</b><br>One valid solution:<br>$E = 8$<br>$A = 9$<br>$T = 2$<br>$H = 7$<br>$P = 1$<br>$L = 0$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
|     |     | <b>BASE + BALL = GAMES</b><br>One valid solution:<br>$B = 7$<br>$A = 4$<br>$S = 8$<br>$E = 3$<br>$L = 5$<br>$G = 1$<br>$M = 9$<br><br><b>LET + LEE = ALL</b><br>One valid solution:<br>$L = 9$<br>$E = 4$<br>$T = 5$<br>$A = 1$<br>(ii) Backtracking & Constraint Propagation (6 Marks)<br>Backtracking is a depth-first search algorithm used to solve CSPs.<br><b>Steps:</b> <ol style="list-style-type: none"> <li>1. Assign a value to a variable.</li> <li>2. Check if constraints are satisfied.</li> <li>3. If violated → backtrack and try another value.</li> <li>4. Continue until solution is found.</li> </ol> In cryptarithmic: <ul style="list-style-type: none"> <li>• Assign digits to letters one by one.</li> <li>• If column addition fails → undo assignment and try another digit.</li> </ul> <b>Constraint Propagation (3 Marks)</b><br>Constraint propagation reduces the domain of variables before or during search.<br><b>Techniques:</b> <ul style="list-style-type: none"> <li>• Forward Checking</li> <li>• Arc Consistency (AC-3)</li> </ul> <b>In Cryptarithmic:</b> <ul style="list-style-type: none"> <li>• If one letter is assigned a digit → remove it from other domains.</li> <li>• Carry constraints restrict possible values early.</li> <li>• Reduces search space significantly.</li> </ul> |     |
|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |     |
| 14. | (a) | i) Explain the process of converting a well-formed formula (WFF) into clause form.<br>(5 Marks)<br>ii) Convert the following formula into clause form step by step:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CO4 |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>a) <math>\forall x: [\text{Leader}(x) \rightarrow \exists y: (\text{Follower}(y) \wedge \text{Supports}(y, x)) \vee \text{Protects}(x, y)]</math></p> <p>b) <math>\forall x [\text{Manager}(x) \rightarrow \exists y (\text{Employee}(y) \wedge \text{Supervises}(x, y))]</math> <span style="float: right;">(8 Marks)</span></p> <p><b>Answer</b></p> <p>Clause form (CNF) is a standard representation of First-Order Logic (FOL) expressions used in <b>resolution-based theorem proving</b>.</p> <p><b>Steps to Convert WFF to Clause Form:</b></p> <p><b>Eliminate Implications and Biconditionals</b></p> <ul style="list-style-type: none"> <li><math>P \rightarrow Q \equiv \neg P \vee Q</math></li> <li><math>P \leftrightarrow Q \equiv (\neg P \vee Q) \wedge (P \vee \neg Q)</math></li> </ul> <p><b>Move NOT (<math>\neg</math>) Inward</b> (Using De Morgan's Laws)</p> <ul style="list-style-type: none"> <li><math>\neg(P \wedge Q) \equiv \neg P \vee \neg Q</math></li> <li><math>\neg(P \vee Q) \equiv \neg P \wedge \neg Q</math></li> <li><math>\neg \forall x P(x) \equiv \exists x \neg P(x)</math></li> <li><math>\neg \exists x P(x) \equiv \forall x \neg P(x)</math></li> </ul> <p><b>Standardize Variables</b></p> <ul style="list-style-type: none"> <li>Rename variables to avoid conflicts.</li> </ul> <p><b>Convert to Prenex Normal Form (PNF)</b></p> <ul style="list-style-type: none"> <li>Move all quantifiers to the front.</li> </ul> <p><b>Skolemization</b></p> <ul style="list-style-type: none"> <li>Remove <math>\exists</math> by introducing: <ul style="list-style-type: none"> <li>Skolem constant (if independent)</li> <li>Skolem function (if dependent)</li> </ul> </li> </ul> <p><b>Drop Universal Quantifiers (<math>\forall</math>)</b></p> <ul style="list-style-type: none"> <li>Assume all variables are universally quantified. <b>Convert to CNF</b></li> <li>Convert to conjunction of disjunctions.</li> </ul> <p><b>Write as Set of Clauses</b></p> <ul style="list-style-type: none"> <li>Each disjunction becomes a separate clause.</li> </ul> <p>(ii) Conversion into Clause Form</p> <p><b>(a) Formula 1</b></p> <p><b>Given:</b></p> <p><math>\forall x [\text{Leader}(x) \rightarrow \exists y ((\text{Follower}(y) \wedge \text{Supports}(y, x)) \vee \text{Protects}(x, y))]</math></p> |  |
|  | <p><b>Step 1: Eliminate Implication</b></p> <p><math>\forall x [\neg \text{Leader}(x) \vee \exists y ((\text{Follower}(y) \wedge \text{Supports}(y, x)) \vee \text{Protects}(x, y))]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |  |
|  | <p><b>Step 2: Move Quantifiers (PNF)</b></p> <p><math>\forall x \exists y [\neg \text{Leader}(x) \vee ((\text{Follower}(y) \wedge \text{Supports}(y, x)) \vee \text{Protects}(x, y))]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |
|  | <p><b>Step 3: Skolemization</b></p> <p>Since <math>\exists y</math> depends on <math>x</math>:</p> <p><math>y = f(x)</math></p> <p><math>\forall x [\neg \text{Leader}(x) \vee ((\text{Follower}(f(x)) \wedge \text{Supports}(f(x), x)) \vee \text{Protects}(x, f(x)))]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |  |
|  | <p><b>Step 4: Drop <math>\forall</math></b></p> <p><math>\neg \text{Leader}(x) \vee ((\text{Follower}(f(x)) \wedge \text{Supports}(f(x), x)) \vee \text{Protects}(x, f(x)))</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |  |
|  | <p><b>Step 5: Convert to CNF (Distribute <math>\vee</math> over <math>\wedge</math>)</b></p> <p><math>(\neg \text{Leader}(x) \vee \text{Follower}(f(x)) \vee \text{Protects}(x, f(x))) \wedge (\neg \text{Leader}(x) \vee \text{Supports}(f(x), x) \vee \text{Protects}(x, f(x)))</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
|  | <p><b>Final Clause Set:</b></p> <ol style="list-style-type: none"> <li><math>\{\neg \text{Leader}(x) \vee \text{Follower}(f(x)) \vee \text{Protects}(x, f(x))\}</math></li> <li><math>\{\neg \text{Leader}(x) \vee \text{Supports}(f(x), x) \vee \text{Protects}(x, f(x))\}</math></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |
|  | <p><b>(b) Formula 2</b></p> <p><b>Given:</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |

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|--|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|  |     | $\forall x[\text{Manager}(x) \rightarrow \exists y(\text{Employee}(y) \wedge \text{Supervises}(x,y))]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
|  |     | <b>Step 1: Eliminate Implication</b><br>$\forall x[\neg \text{Manager}(x) \vee \exists y(\text{Employee}(y) \wedge \text{Supervises}(x,y))]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
|  |     | <b>Step 2: Move Quantifiers (PNF)</b><br>$\forall x \exists y[\neg \text{Manager}(x) \vee (\text{Employee}(y) \wedge \text{Supervises}(x,y))]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
|  |     | <b>Step 3: Skolemization</b><br>Since y depends on x:<br>$y = g(x)$<br>$\forall x[\neg \text{Manager}(x) \vee (\text{Employee}(g(x)) \wedge \text{Supervises}(x,g(x)))]$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |     |
|  |     | <b>Step 4: Drop <math>\forall</math></b><br>$\neg \text{Manager}(x) \vee (\text{Employee}(g(x)) \wedge \text{Supervises}(x,g(x)))$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
|  |     | <b>Step 5: Convert to CNF</b><br>$(\neg \text{Manager}(x) \vee \text{Employee}(g(x))) \wedge (\neg \text{Manager}(x) \vee \text{Supervises}(x,g(x)))$<br><b>Final Clause Set:</b> <ol style="list-style-type: none"> <li><math>\{\neg \text{Manager}(x) \vee \text{Employee}(g(x))\}</math></li> <li><math>\{\neg \text{Manager}(x) \vee \text{Supervises}(x,g(x))\}</math></li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |     |
|  |     | <b>(OR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |     |
|  | (b) | <p>Explain quantifiers in first-order logic and represent the following English sentences in first-order logic form.</p> <p>All programmers know at least one programming language.</p> <p>Every student who studies AI likes at least one AI course.</p> <p>There is a person who knows every student.</p> <p>Not all birds are small</p> <p>Every cat is an animal.</p> <p>Every programmer writes code.</p> <p>Every teacher guides their students.</p> <p>Some teachers are researchers.</p> <p>Every university has a library.</p> <p>There is at least one planet that supports life.</p> <p>Not all students are good at math.</p> <p>There is a book that is on every shelf.</p> <p>Some birds cannot fly.</p> <p>Every researcher publishes at least one paper in some journal.</p> <p>If a number is prime, it is not divisible by any number other than 1 and itself.</p> <p><b>(i) Explain Quantifiers in First-Order Logic (4 Marks)</b></p> <p><b>Quantifiers in First-Order Logic (FOL)</b></p> <p>Quantifiers specify <b>how many objects</b> in the domain satisfy a predicate.</p> <p><b>Universal Quantifier (<math>\forall</math>) – “For All”</b></p> <ul style="list-style-type: none"> <li>Symbol: <math>\forall</math></li> <li>Meaning: Statement is true for every object in the domain.</li> <li>Example:<br/> <math>\forall x \text{ Cat}(x) \rightarrow \text{Animal}(x)</math><br/>           (All cats are animals)</li> </ul> <p><b>Existential Quantifier (<math>\exists</math>) – “There Exists”</b></p> | CO4 |

- Symbol:  $\exists$
- Meaning: There is at least one object that satisfies the predicate.
- Example:  
 $\exists x \text{ Student}(x) \wedge \text{Intelligent}(x)$   
 (There exists an intelligent student)

**Part (ii) Represent the Sentences in First-Order Logic (9 Marks)**

**All programmers know at least one programming language.**

$\forall x [\text{Programmer}(x) \rightarrow \exists y (\text{Language}(y) \wedge \text{Knows}(x,y))]$

**Every student who studies AI likes at least one AI course.**

$\forall x [(\text{Student}(x) \wedge \text{Studies}(x,\text{AI})) \rightarrow \exists y (\text{AICourse}(y) \wedge \text{Likes}(x,y))]$

**There is a person who knows every student.**

$\exists x [\text{Person}(x) \wedge \forall y (\text{Student}(y) \rightarrow \text{Knows}(x,y))]$

**Not all birds are small.**

$\exists x [\text{Bird}(x) \wedge \neg \text{Small}(x)]$

**Every cat is an animal.**

$\forall x [\text{Cat}(x) \rightarrow \text{Animal}(x)]$

**Every programmer writes code.**

$\forall x [\text{Programmer}(x) \rightarrow \text{WritesCode}(x)]$

**Every teacher guides their students.**

$\forall x [\text{Teacher}(x) \rightarrow \forall y (\text{Student}(y) \rightarrow \text{Guides}(x,y))]$

**Some teachers are researchers.**

$\exists x [\text{Teacher}(x) \wedge \text{Researcher}(x)]$

**Every university has a library.**

$\forall x [\text{University}(x) \rightarrow \exists y (\text{Library}(y) \wedge \text{Has}(x,y))]$

**There is at least one planet that supports life.**

$\exists x [\text{Planet}(x) \wedge \text{SupportsLife}(x)]$

**Not all students are good at math.**

$\exists x [\text{Student}(x) \wedge \neg \text{GoodAtMath}(x)]$

**There is a book that is on every shelf.**

$\exists x [\text{Book}(x) \wedge \forall y (\text{Shelf}(y) \rightarrow \text{On}(x,y))]$

**Some birds cannot fly.**

$\exists x [\text{Bird}(x) \wedge \neg \text{Fly}(x)]$

**Every researcher publishes at least one paper in some journal.**

$\forall x [\text{Researcher}(x) \rightarrow \exists y \exists z (\text{Paper}(y) \wedge \text{Journal}(z) \wedge \text{Publishes}(x,y) \wedge \text{In}(y,z))]$

**If a number is prime, it is not divisible by any number other than 1 and itself.**

$\forall x [\text{Prime}(x) \rightarrow \forall y ((\text{Divides}(y,x) \wedge y \neq 1 \wedge y \neq x) \rightarrow \text{False})]$

OR more clearly:

$\forall x [\text{Prime}(x) \rightarrow \forall y ((\text{Divides}(y,x)) \rightarrow (y = 1 \vee y = x))]$

|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |     |
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| 15. | (a) | <p>i) Define Ontological Engineering and explain its significance in AI knowledge representation. (6 Marks)</p> <p>ii) Discuss how reasoning systems for categories help in AI and provide an example of their application. (7 Marks)</p> <p>Ontological Engineering is the process of designing, developing, and maintaining ontologies, which define the concepts, relationships, and rules within a domain. In AI, it helps structure knowledge so that machines can understand, process, and infer information efficiently.</p> <p>An ontology is a structured representation of knowledge in a specific domain, including:</p> <ul style="list-style-type: none"> <li>• Concepts (Entities) – Objects, people, events, etc.</li> <li>• Relationships – How concepts are related.</li> <li>• Properties (Attributes) – Characteristics of concepts.</li> <li>• Rules &amp; Constraints – Logical conditions for inference.</li> </ul> <p>2. Significance of Ontological Engineering in AI</p> <p>Standardized Knowledge Representation – Ensures AI systems share and understand knowledge consistently.</p> <p>Improves Reasoning &amp; Inference – Enables AI to derive new facts from existing knowledge.</p> <p>Enhances Interoperability – Different AI systems can communicate using shared ontologies.</p> <p>Facilitates Machine Learning &amp; NLP – Helps AI understand human language more accurately.</p> <p>Used in Various AI Applications – Semantic web, expert systems, robotics, and decision support systems.</p> <p>3. Example of Ontology in AI</p> <p>Medical Diagnosis Ontology</p> <ul style="list-style-type: none"> <li>• Concepts: Disease, Symptoms, Treatments, Patients.</li> <li>• Relationships: A disease has symptoms; a treatment cures a disease.</li> <li>• Properties: Fever (Symptom) → Malaria (Disease).</li> <li>• Rule: IF Fever &amp; Chills &amp; Mosquito Bite → THEN Malaria is likely.</li> </ul> <p>AI can use this ontology to infer diagnoses, helping doctors in medical decision-making!</p> <p>(ii) Reasoning Systems for Categories in AI (7 Marks)</p> <p>1. What Are Reasoning Systems for Categories?</p> <p>A reasoning system for categories allows AI to classify and infer knowledge by organizing information into hierarchical categories. AI can generalize, specialize, and relate concepts based on these categories.</p> <p>Category-Based Reasoning Helps AI To:</p> <ul style="list-style-type: none"> <li>• Classify objects &amp; concepts (e.g., "All dogs are animals").</li> <li>• Perform inheritance-based reasoning (e.g., "A Labrador inherits properties of dogs").</li> <li>• Make logical inferences based on category hierarchies.</li> </ul> <p>2. How Category-Based Reasoning Works</p> <p>Example: Animal Classification in AI</p> <ul style="list-style-type: none"> <li>• Top-Level Category: Animal <ul style="list-style-type: none"> <li>◦ Subcategory: Mammal <ul style="list-style-type: none"> <li>▪ Subcategory: Dog <ul style="list-style-type: none"> <li>▪ Instance: Labrador</li> </ul> </li> </ul> </li> </ul> </li> </ul> <p>Reasoning:</p> <ul style="list-style-type: none"> <li>• Since a Labrador is a Dog, it inherits the properties of Mammals.</li> <li>• AI can infer: Labradors breathe oxygen, have fur, and are warm-blooded.</li> <li>• If AI learns that "Mammals give birth to live young", it can infer that dogs do too.</li> </ul> <p>This method allows AI to apply general knowledge to specific cases!</p> | CO5 |
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|  |     | <p>3. Application Example: AI Chatbots in Customer Support</p> <ul style="list-style-type: none"> <li>A banking AI assistant categorizes customer queries: <ul style="list-style-type: none"> <li>Category: Account Issues <ul style="list-style-type: none"> <li>Subcategory: Password Reset</li> <li>Subcategory: Balance Inquiry</li> </ul> </li> </ul> </li> </ul> <p>How Reasoning Works:</p> <ul style="list-style-type: none"> <li>If a user asks, "I forgot my login details," AI recognizes it as a Password Reset issue.</li> <li>AI retrieves appropriate solutions based on category-based reasoning.</li> </ul> <p>By structuring knowledge into categories, AI efficiently understands and responds to queries</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |     |
|  |     | (OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
|  | (b) | <p>i) Discuss the importance of mental objects in Artificial Intelligence and explain how they contribute to reasoning processes in intelligent systems. (7 Marks)</p> <p>ii) Explain the basic principles of modal logic and describe how it is used to represent uncertainty, beliefs, or knowledge in AI systems. Support your answer with a suitable example. (6 Marks)</p> <p>Mental objects are internal representations of knowledge that AI systems use to reason, infer, and make decisions. These objects include beliefs, intentions, goals, desires, and perceptions that help AI simulate human-like thought processes.</p> <p>2. Significance of Mental Objects in AI</p> <p>Enhancing Decision-Making: AI can model beliefs and intentions to make informed decisions.</p> <p>Simulating Human Cognition: AI systems use mental objects to predict behaviors and respond intelligently.</p> <p>Enabling Context-Aware Reasoning: AI adapts its responses based on its belief state.</p> <p>Facilitating Multi-Agent Systems: Mental objects help AI understand and predict other agents' actions.</p> <p>3. Role of Mental Objects in Reasoning Systems</p> <p>AI systems use mental objects in three key ways:</p> <p>Belief-Based Reasoning: AI maintains a belief state to store and update knowledge.</p> <p>Goal-Oriented Planning: AI uses mental objects like intentions to achieve long-term objectives.</p> <p>Handling Uncertainty: AI adjusts its knowledge when new information contradicts prior beliefs.</p> <p>4. Example: AI Personal Assistant</p> <p>An AI assistant like Siri or Alexa maintains mental objects:</p> <ul style="list-style-type: none"> <li>Belief: "User prefers morning news updates."</li> <li>Intention: "Provide news every day at 7 AM."</li> <li>Perception Update: "User asked for sports news."</li> <li>Adaptation: AI updates its knowledge to prioritize sports news in the morning.</li> </ul> <p>By using mental objects, AI improves its reasoning and user experience!</p> <p>(ii) Fundamentals of Modal Logic and Its Role in AI (6 Marks)</p> <p>1. What Is Modal Logic?</p> <p>Modal logic extends classical logic by introducing modes of truth, such as:</p> <ul style="list-style-type: none"> <li>Necessity : A statement is necessarily true in all possible worlds.</li> <li>Possibility : A statement is possibly true in at least one possible world.</li> </ul> | CO5 |

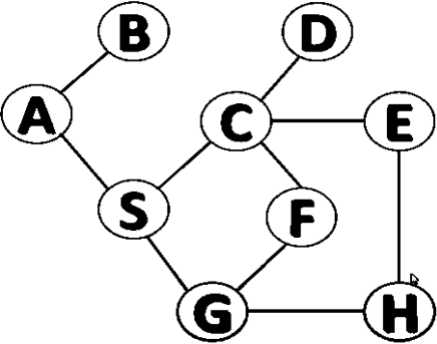
|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
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|  | <p>Modal logic represents beliefs, knowledge, uncertainty, and hypothetical reasoning.</p> <p>2. How Modal Logic Helps in AI</p> <p>Modeling Uncertainty: AI can reason about what may or may not be true.</p> <p>Representing Beliefs &amp; Knowledge: AI can differentiate between what it knows and what it assumes.</p> <p>Handling Hypothetical Scenarios: AI can plan actions based on possible future outcomes.</p> <p>Improving Decision-Making: AI agents consider different possibilities before making a decision.</p> <p>3. Example: AI in Self-Driving Cars</p> <p>A self-driving car uses modal logic to reason about possible situations:</p> <p>Belief: "The traffic light is red." → (Necessarily true)</p> <p>Uncertainty: "The pedestrian might cross the road." → (Possibly true)</p> <p>Action Based on Modal Logic:</p> <ul style="list-style-type: none"> <li>• If (light is red), then stop.</li> <li>• If (pedestrian may cross), then slow down to be cautious.</li> </ul> <p>Modal logic helps AI handle uncertainty and improve safety!</p> <p>Mental objects enable AI to store, update, and use knowledge for intelligent reasoning.</p> <ul style="list-style-type: none"> <li>• Modal logic provides a powerful framework for handling uncertainty, beliefs, and hypothetical scenarios.</li> <li>• Both concepts are crucial in AI decision-making, robotics, multi-agent systems, and knowledge representation</li> </ul> |  |
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**PART C**  
(Case study/Comprehensive type Questions)

**(1 x 15 = 15 marks)**

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| 16. | (a) | <p>i) Apply BFS and DFS to the following State Space Graph after constructing a Search Tree, Where S is the Initial Node, and G is the Goal Node: (8 Marks)</p> <pre> graph TD     S((S)) --&gt; d((d))     S((S)) --&gt; p((p))     d((d)) --&gt; a((a))     d((d)) --&gt; e((e))     a((a)) --&gt; b((b))     b((b)) --&gt; c((c))     c((c)) --&gt; e((e))     e((e)) --&gt; f((f))     e((e)) --&gt; h((h))     e((e)) --&gt; r((r))     f((f)) --&gt; G((G))     h((h)) --&gt; p((p))     h((h)) --&gt; q((q))     p((p)) --&gt; q((q)) </pre> <p>Answer:<br/>For BFS, Path: SERFG<br/>For DFS, Path: SDERFG</p> <p>ii) Apply BFS and DFS to the following State Space Graph after constructing a Search Tree <b>Where A is the Initial Node and H is the Goal Node: (7 Mark)</b></p> | CO1       |



|  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |     |
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|  |     |  <p>BFS:A-S-G-H<br/>DFS-A-S-G-H</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
|  |     | (OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |     |
|  | (b) | <p>i) Develop a Python program to implement an AI agent for the vacuum-cleaner world. (5 Marks)</p> <p>ii) Solve the Water Jug Problem:<br/>You are given two jugs — one with a capacity of 4 gallons and the other with 3 gallons. There are no measurement markings on either jug. A pump is available to fill the jugs with water.<br/>Using the following assumptions:<br/>A jug can be filled completely from the pump.<br/>Water can be emptied onto the ground.<br/>Water can be poured from one jug to the other until either the source jug is empty or the destination jug is full.<br/>No other measuring devices are available.<br/>Explain how you can obtain exactly 2 gallons of water in the 4-gallon jug. (10 Marks)</p> <pre> I)import random  class VacuumCleaner:     def __init__(self, grid_size=2):         self.grid_size = grid_size         self.environment = [[random.choice(["clean", "dirty"]) for _ in range(grid_size)] for _ in range(grid_size)]         self.position = (0, 0) # Starts at top-left corner      def display_environment(self):         for row in self.environment:             print(row)         print("Vacuum position:", self.position)      def clean(self):         x, y = self.position         if self.environment[x][y] == "dirty":             self.environment[x][y] = "clean"             print(f"Cleaned at position ({x}, {y})")      def move(self):         x, y = self.position         if y + 1 &lt; self.grid_size:             self.position = (x, y + 1) # Move Right         elif x + 1 &lt; self.grid_size:             self.position = (x + 1, y) # Move Down         else: </pre> | CO1 |

|  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
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|  |  | <pre> print("Cleaning Completed")  def run(self):     print("Initial Environment:")     self.display_environment()     while any("dirty" in row for row in self.environment):         self.clean()         self.move()     print("Final Environment:")     self.display_environment()  # Run the Vacuum Cleaner vacuum = VacuumCleaner() vacuum.run() </pre> <p><b>II) Step 1: Problem Formulation (3 Marks)</b></p> <p><b>State Representation</b><br/> Represent each state as:<br/> (x,y)<br/> Where:</p> <ul style="list-style-type: none"> <li>• x = amount of water in 4-gallon jug</li> <li>• y = amount of water in 3-gallon jug</li> </ul> <p>Initial State:<br/> (0,0)<br/> Goal State:<br/> (2,y)<br/> (We need exactly 2 gallons in the 4-gallon jug)</p> <p><b>Possible Actions</b></p> <ul style="list-style-type: none"> <li>• Fill 4-gallon jug</li> <li>• Fill 3-gallon jug</li> <li>• Empty 4-gallon jug</li> <li>• Empty 3-gallon jug</li> <li>• Pour 4 → 3</li> <li>• Pour 3 → 4</li> </ul> <p><b>Step 2: Solution Steps (5 Marks)</b><br/> Start from <b>(0,0)</b><br/> Fill 3-gallon jug<br/> → (0,3)<br/> Pour 3 → 4<br/> → (3,0)<br/> Fill 3-gallon jug again<br/> → (3,3)<br/> Pour 3 → 4 until 4-gallon jug is full<br/> (4-gallon jug needs only 1 more gallon)<br/> → (4,2)<br/> Empty 4-gallon jug<br/> → (0,2)<br/> Pour remaining 2 gallons from 3 → 4<br/> → (2,0) - Goal Achieved</p> <p><b>Final State:</b><br/> (2,0)<br/> Exactly <b>2 gallons in the 4-gallon jug.</b></p> |  |
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|  |  | <b>Step 3: Explanation (2 Marks)</b><br>This problem is solved using <b>state space search</b> , where: <ul style="list-style-type: none"> <li>• Each state represents water amounts.</li> <li>• Actions move from one state to another.</li> <li>• The goal state is reached systematically.</li> </ul> This can be solved using: <ul style="list-style-type: none"> <li>• Breadth-First Search (BFS)</li> <li>• Depth-First Search (DFS)</li> </ul> |  |
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|                  | For Set 1 QP |       |       |       |       |       |       |       |       |       |          |       |
|------------------|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|-------|
|                  | Part - A     |       |       |       |       |       |       |       |       |       |          |       |
| Question No.     | 1            | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |          |       |
| Knowledge Level  | K2           | K1    | K1    | K4    | K2    | K2    | K1    | K3    | K1    | K2    |          |       |
| Difficulty Level | 1            | 1     | 2     | 1     | 2     | 1     | 1     | 2     | 1     | 1     |          |       |
|                  | Part - B     |       |       |       |       |       |       |       |       |       | Part - C |       |
| Question No.     | 11(a)        | 11(b) | 12(a) | 12(b) | 13(a) | 13(b) | 14(a) | 14(b) | 15(a) | 15(b) | 16(a)    | 16(b) |
| Knowledge Level  | K2           | K2    | K3    | K4    | K3    | K4    | K3    | K4    | K4    | K4    | K4       | K4    |
| Difficulty Level | 4            | 4     | 4     | 3     | 4     | 4     | 4     | 4     | 4     | 4     | 4        | 4     |

SET 2

QP Setter Name / Dept:

QP Code:

Reg. No

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**SAVEETHA ENGINEERING COLLEGE**

(Autonomous, Affiliated to Anna University, Chennai)

B.E./B.Tech. End Semester Examinations – Apr/May 2026 (R2019/R2024)

Common to AI/ML, AI/DS, CSE, IT (Branches)

**19AI405/CS3405-FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**

Time: Three hours

Maximum marks: 100

Answer All Questions

**PART A**

**(10 x 2 = 20 marks)**

|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO        |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 1. | <p>List out the characteristics of an intelligent agent.</p> <p>Internal characteristics are</p> <ul style="list-style-type: none"> <li>– Learning/reasoning: an agent has the ability to learn from previous experience and to successively adapt its own behavior to the environment.</li> <li>– Reactivity: an agent must be capable of reacting appropriately to influences or information from its environment.</li> <li>– Autonomy: an agent must have both control over its actions and internal states. The degree of the agent's autonomy can be specified. There may need intervention from the user only for important decisions.</li> <li>– Goal-oriented: an agent has well-defined goals and gradually influence its environment and so achieve its own goals.</li> </ul> <p>External characteristics are</p> <ul style="list-style-type: none"> <li>– Communication: an agent often requires an interaction with its environment to fulfill its tasks, such as human, other agents, and arbitrary information sources.</li> <li>– Cooperation: cooperation of several agents permits faster and better solutions for complex tasks that exceed the capabilities of a single agent.</li> <li>– Mobility: an agent may navigate within electronic communication networks.</li> <li>– Character: like human, an agent may demonstrate an external behavior with many human characters as possible.</li> </ul> | CO<br>CO1 |
| 2. | <p>Differentiate between a fully observable and a partially observable environment.</p> <p><b>Fully Observable Environment</b></p> <ul style="list-style-type: none"> <li>• Agent has <b>complete access</b> to the environment's state at each time.</li> <li>• No hidden information; decisions are made with <b>certainty</b>.</li> <li>• Easier to design optimal agents since the world state is fully known.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CO1       |

|                                                                   | <ul style="list-style-type: none"><li>• <b>Example:</b> Chess, Tic-tac-toe, solving a maze with full map visible.</li></ul> <b>Fully observable</b> = complete knowledge, deterministic reasoning possible.<br><b>Partially Observable Environment</b> <ul style="list-style-type: none"><li>• Agent has <b>limited or incomplete access</b> to the environment's state.</li><li>• Some aspects are hidden or uncertain; requires <b>estimation, belief states, or probability</b>.</li><li>• More complex; agents must deal with <b>ambiguity and uncertainty</b>.</li><li>• <b>Example:</b> Poker, self-driving cars (limited sensor view), medical diagnosis.</li></ul> <b>Partially observable</b> = incomplete knowledge, probabilistic/approximate reasoning required.                                                                                              |              |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
|-------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------------|---------------------------------------------|------------------------------------------------------------------|-----------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------|-------------------------------------------------------|-----|
| 3.                                                                | List any four disadvantages of the Hill Climbing algorithm.<br><b>Gets Stuck in Local Maximum</b> <ul style="list-style-type: none"><li>• <b>Stops at a local optimum which is not the global optimum.</b></li></ul> <b>Plateau Problem</b> <ul style="list-style-type: none"><li>• <b>When many neighboring states have the same value, the algorithm cannot decide where to move.</b></li></ul> <b>Ridge Problem</b> <ul style="list-style-type: none"><li>• <b>When the search space has a ridge structure, the algorithm may oscillate or move very slowly.</b></li></ul> <b>Not Complete</b> <ul style="list-style-type: none"><li>• <b>It may fail to find a solution even if one exists.</b></li></ul>                                                                                                                                                             | CO2          |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
| 4.                                                                | State the property of an admissible heuristic.<br><b>An admissible heuristic is one that never overestimates the true cost of reaching the goal state.</b><br><b>Property:</b><br>For every node $n$ ,<br>$h(n) \leq h^*(n)$<br><b>Where:</b> <ul style="list-style-type: none"><li>• <math>h(n)</math> = heuristic estimate</li><li>• <math>h^*(n)</math> = actual minimum cost to reach the goal</li></ul> <b>An admissible heuristic is optimistic (it may underestimate, but never overestimate).</b><br><br><b>If a heuristic is admissible, the A* search algorithm is optimal (guaranteed to find the least-cost solution).</b>                                                                                                                                                                                                                                    | CO2          |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
| 5.                                                                | What is a Hamiltonian graph?<br>A graph that contains a <b>Hamiltonian cycle</b> , i.e., a cycle that <b>visits every vertex exactly once</b> and returns to the starting vertex. <ul style="list-style-type: none"><li>• A <b>Hamiltonian path</b> is a path that visits every vertex exactly once (not required to return).</li><li>• If such a cycle exists → the graph is called <b>Hamiltonian</b>.</li><li>• Unlike Eulerian graphs (which focus on edges), <b>Hamiltonian graphs focus on visiting all vertices</b>.</li></ul> <b>Example:</b> <ul style="list-style-type: none"><li>• A simple cycle graph <math>C_n</math> with <math>n \geq 3</math> is Hamiltonian.</li><li>• A complete graph <math>K_n</math> (for <math>n \geq 3</math>) is always Hamiltonian.</li></ul> <b>Hamiltonian Graph</b> = Graph with a cycle visiting all vertices exactly once. | CO3          |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
| 6.                                                                | Differentiate between Forward Checking and Backtracking in CSP. <table><thead><tr><th>Backtracking</th><th>Forward Checking</th></tr></thead><tbody><tr><td>Basic depth-first search technique for CSP.</td><td>An improved version of backtracking with constraint propagation.</td></tr><tr><td>Assigns values to variables one by one.</td><td>After assignment, reduces domains of future variables.</td></tr><tr><td>Checks constraints only after full assignment or conflict occurs.</td><td>Checks constraints immediately after each assignment.</td></tr></tbody></table>                                                                                                                                                                                                                                                                                       | Backtracking | Forward Checking | Basic depth-first search technique for CSP. | An improved version of backtracking with constraint propagation. | Assigns values to variables one by one. | After assignment, reduces domains of future variables. | Checks constraints only after full assignment or conflict occurs. | Checks constraints immediately after each assignment. | CO3 |
| Backtracking                                                      | Forward Checking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
| Basic depth-first search technique for CSP.                       | An improved version of backtracking with constraint propagation.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |              |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
| Assigns values to variables one by one.                           | After assignment, reduces domains of future variables.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |              |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |
| Checks constraints only after full assignment or conflict occurs. | Checks constraints immediately after each assignment.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |                  |                                             |                                                                  |                                         |                                                        |                                                                   |                                                       |     |

|     | May explore many invalid paths.<br>Less efficient.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Prunes invalid values early, reducing search space.<br>More efficient than simple backtracking. |     |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------|-----|-------------------|---|---|---|---|---|---|---|---|---|---|---|---|--|--|
| 7.  | Define the logical connective for implication. Provide its symbol and truth table. <ul style="list-style-type: none"><li>• <b>Definition:</b> The implication <math>P \rightarrow Q</math> means “If <math>P</math>, then <math>Q</math>”.</li><li>• It is <b>false only when P is true and Q is false</b>, otherwise true.</li><li>• <b>Symbol:</b> <math>\rightarrow</math> (sometimes written as <math>\Rightarrow</math>).</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                 | CO4 |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
|     | <b>Truth Table for Implication</b> <table><tr><th>P</th><th>Q</th><th><math>P \rightarrow Q</math></th></tr><tr><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>T</td></tr><tr><td>F</td><td>F</td><td>T</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | P                                                                                               | Q   | $P \rightarrow Q$ | T | T | T | T | F | F | F | T | T | F | F | T |  |  |
| P   | Q                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | $P \rightarrow Q$                                                                               |     |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| T   | T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | T                                                                                               |     |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| T   | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | F                                                                                               |     |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| F   | T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | T                                                                                               |     |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| F   | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | T                                                                                               |     |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| 8.  | Define resolution in logic.<br><b>Resolution is an inference rule used to derive a new clause by eliminating complementary literals.</b><br><b>If</b><br><b>(<math>P \vee Q</math>)</b><br><b>(<math>\neg P \vee R</math>)</b><br><b>Then by resolution:</b><br><b>(<math>Q \vee R</math>)</b><br><b>Used in automated theorem proving and logical reasoning.</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                 | CO4 |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| 9.  | Define ontological engineering and its significance in knowledge representation. <ul style="list-style-type: none"><li>• Ontological engineering is the process of <b>designing, building, and maintaining ontologies</b>—formal, structured representations of concepts and their relationships within a domain.</li><li>• An <b>ontology</b> defines the entities, categories, properties, and relations that exist in a particular area of knowledge.</li><li>• <b>Ontological engineering</b> is about creating structured knowledge models (ontologies), and it is significant because it enables <b>clear, reusable, and sharable knowledge representation</b> for intelligent systems.</li></ul><br><b>Significance in Knowledge Representation</b> <ol style="list-style-type: none"><li>1. <b>Provides a common vocabulary</b> – ensures consistent meaning of terms across systems.</li><li>2. <b>Supports interoperability</b> – enables different agents/systems to share and reuse knowledge.</li><li>3. <b>Facilitates reasoning</b> – allows logical inference over structured domain knowledge.</li><li>4. <b>Domain modeling</b> – captures expert knowledge in an explicit, reusable way.</li><li>5. <b>Foundation for AI systems</b> – essential in areas like semantic web, natural language understanding, and expert systems.</li></ol> |                                                                                                 | CO5 |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |
| 10. | Define STRIPS in the context of classical planning.<br>STRIPS stands for Stanford Research Institute Problem Solver.<br>It is a formal language for describing planning problems in Artificial Intelligence.<br>A planning problem in STRIPS is defined using: <ol style="list-style-type: none"><li>1. Initial State – facts true at the start.</li><li>2. Goal State – conditions that must be satisfied.</li><li>3. Actions (Operators) – each action has:<ul style="list-style-type: none"><li>○ Preconditions: what must be true to apply the action.</li><li>○ Add list: facts that become true after the action.</li></ul></li></ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                 | CO5 |                   |   |   |   |   |   |   |   |   |   |   |   |   |  |  |

|  |                                                                                                                  |  |
|--|------------------------------------------------------------------------------------------------------------------|--|
|  | <ul style="list-style-type: none"> <li>○ Delete list: facts that are no longer true after the action.</li> </ul> |  |
|--|------------------------------------------------------------------------------------------------------------------|--|

## PART B

(5 x 13 = 65 marks)

|    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO      |
|----|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
| 11 | (a) | <p>Describe in detail the different types of Intelligent agents with a neat block diagram and pseudo-code.</p> <p><b>Introduction (2 Marks)</b><br/> An <b>Intelligent Agent</b> is an entity that:</p> <ul style="list-style-type: none"> <li>• <b>Perceives</b> through sensors</li> <li>• <b>Acts</b> through actuators</li> <li>• Aims to maximize performance measure</li> </ul> <p>Agent = Sensors → Agent Program → Actuators</p> <p><b>Types of Intelligent Agents</b></p> <p><b>1. Simple Reflex Agent (3 Marks)</b><br/> <b>Concept:</b></p> <ul style="list-style-type: none"> <li>• Acts only based on current percept</li> <li>• Uses condition–action rules</li> <li>• No memory</li> </ul> <p><b>Block Diagram:</b><br/> Environment → Sensors → [Condition-Action Rules] → Actuators → Environment</p> <p><b>Pseudo-Code:</b><br/> function SIMPLE-REFLEX-AGENT(percept)<br/>   state ← interpret(percept)<br/>   rule ← rule-match(state)<br/>   action ← rule-action(rule)<br/>   return action</p> <p><b>Example:</b><br/> Thermostat system</p> <p><b>2. Model-Based Reflex Agent (2 Marks)</b><br/> <b>Concept:</b></p> <ul style="list-style-type: none"> <li>• Maintains internal state</li> <li>• Handles partially observable environments</li> </ul> <p><b>Block Diagram:</b><br/> Environment → Sensors → [Internal State + Model] → Rules → Actuators</p> <p><b>Pseudo-Code:</b><br/> function MODEL-BASED-AGENT(percept)<br/>   state ← update-state(state, percept)<br/>   rule ← rule-match(state)<br/>   action ← rule-action(rule)<br/>   return action</p> <p><b>3. Goal-Based Agent (2 Marks)</b><br/> <b>Concept:</b></p> <ul style="list-style-type: none"> <li>• Has explicit goals</li> <li>• Uses search/planning to achieve goals</li> </ul> <p><b>Block Diagram:</b><br/> Percept → State → Goal Information → Search/Planning → Action</p> <p><b>Pseudo-Code:</b><br/> function GOAL-BASED-AGENT(percept)<br/>   state ← update-state(state, percept)</p> | CO<br>1 |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |         |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|     | <pre>action ← search(state, goal) return action</pre> <p><b>4. Utility-Based Agent (2 Marks)</b><br/><b>Concept:</b></p> <ul style="list-style-type: none"><li>• Uses utility function</li><li>• Chooses action with highest expected utility</li></ul> <p><b>Block Diagram:</b><br/>Percept → State → Utility Function → Best Action → Actuator</p> <p><b>Pseudo-Code:</b><br/>function UTILITY-BASED-AGENT(percept)<br/>  state ← update-state(state, percept)<br/>  action ← argmax_a Utility(result(state, a))<br/>  return action</p> <p><b>5. Learning Agent (2 Marks)</b><br/><b>Concept:</b><br/>Improves performance from experience.</p> <p><b>Components:</b></p> <ul style="list-style-type: none"><li>• Performance element</li><li>• Learning element</li><li>• Critic</li><li>• Problem generator</li></ul> <p><b>Block Diagram:</b><br/>Environment<br/>↓<br/>Performance Element → Action<br/>↑<br/>Learning Element ← Critic<br/>↑<br/>Problem Generator</p> <p><b>Pseudo-Code:</b><br/>function LEARNING-AGENT(percept)<br/>  action ← performance-element(percept)<br/>  feedback ← critic(percept)<br/>  update-learning(feedback)<br/>  return action</p> |         |
|     | (OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |         |
| (b) | <p>i) Explain how Bidirectional Search works and write its pseudo code. (10 Marks)</p> <p>ii) List the benefits and disadvantages of Bidirectional Search. (3 Marks)</p> <p>Bidirectional search is an algorithm used in AI to find the shortest path between an initial state and a goal state. It simultaneously searches forward from the initial state and backward from the goal state, stopping when the two searches meet in the middle.</p> <p>How Bidirectional Search Works</p> <ol style="list-style-type: none"><li>1. Forward Search: Starts from the initial state and expands outward.</li><li>2. Backward Search: Starts from the goal state and expands outward.</li><li>3. Meeting Point: The search stops when the two fronts meet at a common node, meaning a path has been found.</li></ol> <p>Advantages</p> <ul style="list-style-type: none"><li>• More efficient than unidirectional search in some cases.</li><li>• Reduces search depth, leading to fewer nodes explored.</li><li>• Can be faster in large state spaces.</li></ul>                                                                                                                   | CO<br>1 |



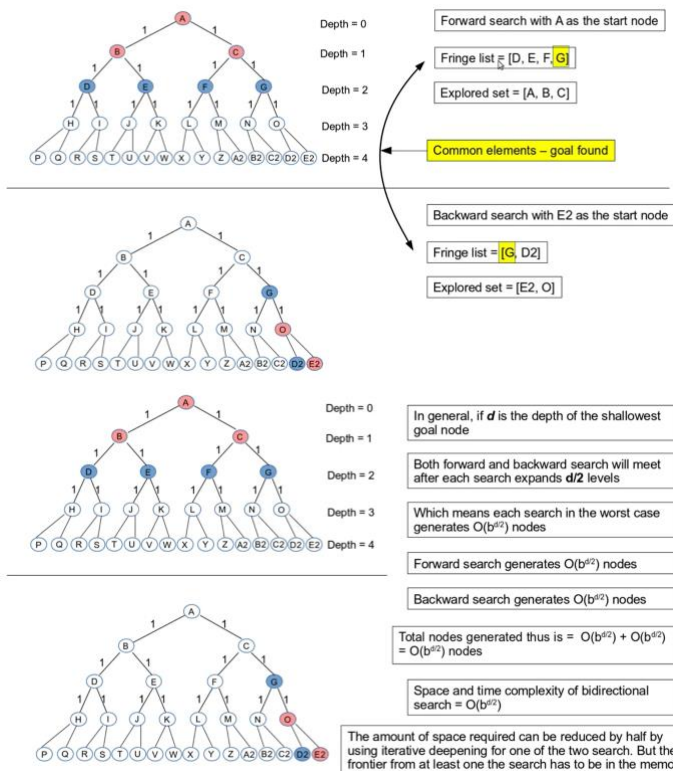
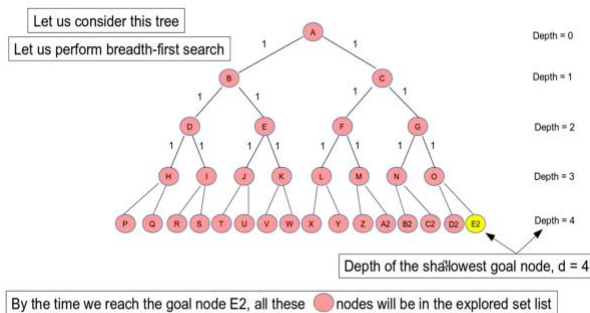
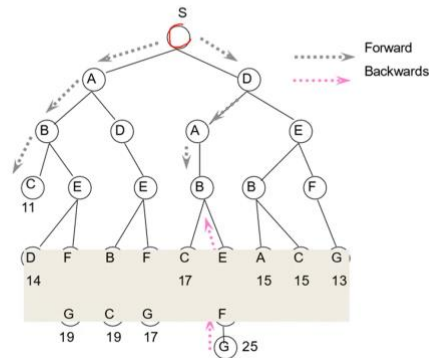
## Disadvantages

- Requires a way to generate predecessor states for backward search.
- Increased memory usage due to maintaining two search trees.
- Harder to implement in cases with multiple goal states.

## Example

- Finding the shortest path in a graph.
- Solving puzzles like the 8-puzzle problem.
- Route planning in navigation systems.

## Bi-directional Search...



Breadth-First Search (BFS) has some advantages over Bidirectional Search in certain scenarios:

1. Simplicity – BFS is straightforward to implement, whereas bidirectional search requires careful management of two simultaneous searches and meeting points.

|    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
|----|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|    |     | <ol style="list-style-type: none"> <li>2. Guaranteed Optimality (for Unweighted Graphs) – BFS finds the shortest path in an unweighted graph, while bidirectional search may require extra handling to ensure optimality.</li> <li>3. Less Overhead in Implementation – BFS only requires a single search tree, whereas bidirectional search needs two trees and a mechanism to detect when they meet.</li> <li>4. Works in All Scenarios – Bidirectional search is effective when the goal state is known and efficiently reachable. However, in problems where the goal is unknown or dynamically changes, BFS is more practical.</li> <li>5. Better for Dense Graphs – In dense graphs with many connections, bidirectional search may not always perform better than BFS due to the high branching factor.</li> </ol> <p>ii) Describe the advantages and limitations of using bidirectional search</p> <p>Advantages of Bidirectional Search</p> <ol style="list-style-type: none"> <li>1. Faster Search Time – By simultaneously searching from the start and goal nodes, bidirectional search can significantly reduce the search space, leading to faster solutions compared to unidirectional searches like BFS or DFS.</li> <li>2. Lower Memory Usage – Since each search (forward and backward) explores fewer nodes than a full BFS, bidirectional search can require less memory in large graphs.</li> <li>3. More Efficient for Large Graphs – In an ideal case, bidirectional search explores roughly <math>O(b^{(d/2)})</math> nodes instead of <math>O(b^d)</math> in a standard BFS (where <math>b</math> is the branching factor and <math>d</math> is the depth of the solution). This makes it much more efficient for deep graphs.</li> <li>4. Guaranteed Optimality in Unweighted Graphs – If BFS is used in both directions, bidirectional search guarantees finding the shortest path in an unweighted graph.</li> <li>5. Works Well for Problems with a Known Goal State – If the goal node is clearly defined, bidirectional search can effectively reduce computation.</li> </ol> <p>Limitations of Bidirectional Search</p> <ol style="list-style-type: none"> <li>1. Complex Implementation – Managing two simultaneous searches, ensuring they meet at a common node, and handling different search strategies require additional overhead compared to BFS or DFS.</li> <li>2. Requires a Defined Goal State – Unlike BFS, which can explore all possible solutions, bidirectional search is only practical when the goal state is known in advance.</li> <li>3. Synchronization Issues – The two search processes need to meet at a common node, which requires extra logic to detect and properly handle the intersection.</li> <li>4. More Memory Intensive in Some Cases – While it generally uses less memory than BFS, if the branching factor is high, both searches might still expand a large number of nodes, leading to significant memory usage.</li> <li>5. Not Always Suitable for Weighted Graphs – If the graph has weights, additional techniques (like bidirectional A*) are required to ensure optimality, which increases complexity</li> </ol> |         |
| 12 | (a) | <p>i) Implement the A* algorithm and explain how it uses a heuristic function to find the optimal path. (7 marks)</p> <p>ii) Discuss the properties and admissibility of the A* algorithm. (6 marks)</p> <p><b>(i) Implement the A Algorithm &amp; Explanation (7 Marks)*</b></p> <p><b>A Algorithm Idea*</b></p> <ul style="list-style-type: none"> <li>• A* is a <b>graph search algorithm</b> that finds the <b>shortest/optimal path</b> from a start node to a goal node.</li> <li>• It uses a <b>heuristic function (h(n))</b> to estimate the cost of the cheapest path from node n to the goal.</li> <li>• It balances: <ul style="list-style-type: none"> <li>◦ <b>g(n)</b>: actual cost from start to node n</li> </ul> </li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CO<br>2 |

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|  |     | <ul style="list-style-type: none"> <li>○ <b>h(n):</b> estimated cost from n to goal</li> <li>• The evaluation function is:<br/> <math>f(n)=g(n)+h(n)</math></li> </ul> <p><b>A* Search Algorithm :</b></p> <ol style="list-style-type: none"> <li>1. Start with OPEN containing only the initial node. Set that node's g value to 0, its h' value to whatever it is, and its f' value to h' + 0, or h'. Set CLOSED to the empty list.</li> <li>2. Until a goal node is found, repeat the following procedure: If there are no nodes on OPEN. report failure. Otherwise, pick the node on OPEN with the lowest f' value. Call it BESTNODE. Remove it from OPEN. Place it on CLOSED. See if BESTNODE is a goal node. If so, exit and report a solution. Otherwise, generate the successors of BESTNODE. For each such SUCCESSOR, do the following: <ol style="list-style-type: none"> <li>a) Set SUCCESSOR to point back to BESTNODE. These backwards links will make it possible to recover the path once a solution is found.</li> <li>b) Compute <math>g(\text{SUCCESSOR}) = g(\text{BESTNODE}) + \text{the cost of getting from BESTNODE to SUCCESSOR}</math>.</li> <li>c) Compute <math>f(\text{SUCCESSOR}) = g(\text{SUCCESSOR}) + h'(\text{SUCCESSOR})</math></li> <li>d) if SUCCESSOR was not already on either OPEN or CLOSED, then put it on OPEN, and add it to the list of BESTNODE's successors.</li> </ol> </li> </ol> <p><b>How A* Uses Heuristic</b></p> <ul style="list-style-type: none"> <li>• At each step, A* chooses the node with the <b>lowest estimated total cost</b><br/> <math>f(n)=g(n)+h(n)</math></li> <li>• The heuristic h(n) <b>guides the search towards the goal</b> by estimating the distance to it.</li> <li>• If h(n) is well-designed, A* expands far fewer nodes than Uniform Cost Search.</li> </ul> <p><b>Example:</b></p> <ul style="list-style-type: none"> <li>• In a grid, if h(n) is the <b>Manhattan distance</b> to the goal, A* quickly converges to the shortest path.</li> </ul> <p>Draw and explain the 8-puzzle problem using heuristic as an example.</p> <p><b>(ii) Properties &amp; Admissibility of A* (6 Marks)</b></p> <p><b>Properties</b></p> <ol style="list-style-type: none"> <li>1. <b>Complete</b> – A* is guaranteed to find a solution if one exists (with finite branching and positive step cost).</li> <li>2. <b>Optimal</b> – A* finds the least-cost (optimal) path if the heuristic is admissible.</li> <li>3. <b>Time Complexity</b> – Exponential in the worst case, but efficient with a good heuristic.</li> <li>4. <b>Space Complexity</b> – High, as it stores all generated nodes in memory (open + closed lists).</li> </ol> <p><b>Admissibility of A*</b></p> <ul style="list-style-type: none"> <li>• A heuristic h(n) is <b>admissible</b> if it <b>never overestimates</b> the true cost to reach the goal:<br/> <math>h(n) \leq h^*(n) \forall n</math><br/> where h*(n) is the actual cost from n to goal.</li> <li>• If the heuristic is admissible, A* is <b>guaranteed to find an optimal solution</b>.</li> </ul> <p>A* evaluates nodes with <math>f(n)=g(n)+h(n)</math><br/> Uses heuristics to reduce search space.<br/> Properties: <b>Complete, Optimal, Efficient with good h(n).</b><br/> <b>Admissibility</b> ensures optimal solutions because heuristic never overestimates.</p> |                                                   |
|  |     | (OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                   |
|  | (b) | <ol style="list-style-type: none"> <li>i) Implement the Steepest-Ascent Hill Climbing algorithm. Discuss a major problem associated with it and propose a strategy to overcome it. (7 marks)</li> <li>ii) Differentiate between local search algorithms and global search algorithms. Provide two examples of each. (6 marks)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>2</p> <p>With it and</p> <p>examples of ea</p> |

|                       |                                                                       | <p><b>(i) Steepest-Ascent Hill Climbing (7 Marks)</b></p> <ul style="list-style-type: none"><li>• A local search algorithm that starts from a candidate solution and <b>moves to the neighbor with the best improvement</b> in the objective function.</li><li>• Stops when no better neighbor is found.</li></ul> <p><b>Pseudo-code for Steepest-Ascent Hill Climbing</b></p> <pre>function SteepestAscentHillClimbing(problem):<br/>    current ← initial_state(problem)<br/>    loop:<br/>        neighbors ← generate_neighbors(current)<br/>        if neighbors is empty:<br/>            return current<br/>        next ← best(neighbors) # neighbor with highest evaluation<br/>        if value(next) ≤ value(current):<br/>            return current # local maximum reached<br/>        current ← next</pre> <p><b>Major Problem</b></p> <ul style="list-style-type: none"><li>• <b>Getting stuck in local maxima, plateaus, or ridges.</b><ul style="list-style-type: none"><li>◦ The algorithm may stop at a suboptimal solution because no neighbor appears better.</li></ul></li></ul> <p><b>Strategy to Overcome</b></p> <ol style="list-style-type: none"><li>1. <b>Random Restart Hill Climbing</b> – Run hill climbing multiple times from different random starting states and take the best solution.</li><li>2. <b>Simulated Annealing</b> – Occasionally accept worse moves to escape local maxima.</li></ol> <p><b>(ii) Local Search vs Global Search (6 Marks)</b></p> <table><tr><th>Aspect</th><th>Local Search Algorithms</th><th>Global Search Algorithms</th></tr><tr><td>Search space</td><td>Explore from a <b>single current state</b> and look at its neighbors.</td><td>Explore the search space <b>more broadly</b> (often population-based or complete search).</td></tr><tr><td>Knowledge used</td><td>Use only <b>local information</b> (neighborhood).</td><td>Consider the <b>entire search space</b> structure or multiple candidate solutions.</td></tr><tr><td>Efficiency</td><td>Usually faster but may get stuck in <b>local optima</b>.</td><td>Slower but can find the <b>global optimum</b>.</td></tr><tr><td>Optimality</td><td>Not guaranteed to find the best global solution.</td><td>Designed to reach the global solution (or a very close one).</td></tr></table> | Aspect | Local Search Algorithms    | Global Search Algorithms      | Search space          | Explore from a <b>single current state</b> and look at its neighbors. | Explore the search space <b>more broadly</b> (often population-based or complete search). | Knowledge used | Use only <b>local information</b> (neighborhood).   | Consider the <b>entire search space</b> structure or multiple candidate solutions. | Efficiency      | Usually faster but may get stuck in <b>local optima</b> .    | Slower but can find the <b>global optimum</b> .                                        | Optimality | Not guaranteed to find the best global solution. | Designed to reach the global solution (or a very close one). |            |                                                    |                                                              |                 |                               |                                                          |         |  |
|-----------------------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|----------------------------|-------------------------------|-----------------------|-----------------------------------------------------------------------|-------------------------------------------------------------------------------------------|----------------|-----------------------------------------------------|------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------|----------------------------------------------------------------------------------------|------------|--------------------------------------------------|--------------------------------------------------------------|------------|----------------------------------------------------|--------------------------------------------------------------|-----------------|-------------------------------|----------------------------------------------------------|---------|--|
| Aspect                | Local Search Algorithms                                               | Global Search Algorithms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
| Search space          | Explore from a <b>single current state</b> and look at its neighbors. | Explore the search space <b>more broadly</b> (often population-based or complete search).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
| Knowledge used        | Use only <b>local information</b> (neighborhood).                     | Consider the <b>entire search space</b> structure or multiple candidate solutions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
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| Optimality            | Not guaranteed to find the best global solution.                      | Designed to reach the global solution (or a very close one).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
| 13                    | (a)                                                                   | <p>i) Differentiate between adversarial search and conventional search strategies, and explain the concept of game-playing in AI. (6 Marks)</p> <table><tr><th>Aspect</th><th>Standard Search Algorithms</th><th>Adversarial Search Algorithms</th></tr><tr><td>Nature of environment</td><td>Assumes a <b>static, deterministic</b> environment.</td><td>Deals with <b>dynamic, competitive</b> environments with an opponent.</td></tr><tr><td>Objective</td><td>Find a path/solution to reach a <b>goal state</b>.</td><td>Find the <b>best strategy</b> to win or maximize outcome against an adversary.</td></tr><tr><td>Decision-making</td><td>Decisions are based on exploring possible actions and costs.</td><td>Decisions are based on both <b>agent's moves</b> and <b>opponent's counter-moves</b>.</td></tr><tr><td>Examples</td><td>BFS, DFS, A* Search.</td><td>Minimax, Alpha-Beta Pruning.</td></tr><tr><td>Evaluation</td><td>Uses heuristic cost to estimate closeness to goal.</td><td>Uses <b>utility/payoff values</b> to evaluate game outcomes.</td></tr><tr><td>Agents involved</td><td>Single decision-making agent.</td><td>Two or more competing agents (e.g., MAX vs MIN players).</td></tr></table> <p>ii) Explain the Minimax algorithm and Alpha–Beta pruning with a game tree example. (7 Marks)</p> <p>Minimax is a decision-making algorithm used in two-player zero-sum games.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Aspect | Standard Search Algorithms | Adversarial Search Algorithms | Nature of environment | Assumes a <b>static, deterministic</b> environment.                   | Deals with <b>dynamic, competitive</b> environments with an opponent.                     | Objective      | Find a path/solution to reach a <b>goal state</b> . | Find the <b>best strategy</b> to win or maximize outcome against an adversary.     | Decision-making | Decisions are based on exploring possible actions and costs. | Decisions are based on both <b>agent's moves</b> and <b>opponent's counter-moves</b> . | Examples   | BFS, DFS, A* Search.                             | Minimax, Alpha-Beta Pruning.                                 | Evaluation | Uses heuristic cost to estimate closeness to goal. | Uses <b>utility/payoff values</b> to evaluate game outcomes. | Agents involved | Single decision-making agent. | Two or more competing agents (e.g., MAX vs MIN players). | CO<br>3 |  |
| Aspect                | Standard Search Algorithms                                            | Adversarial Search Algorithms                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
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| Decision-making       | Decisions are based on exploring possible actions and costs.          | Decisions are based on both <b>agent's moves</b> and <b>opponent's counter-moves</b> .                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
| Examples              | BFS, DFS, A* Search.                                                  | Minimax, Alpha-Beta Pruning.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
| Evaluation            | Uses heuristic cost to estimate closeness to goal.                    | Uses <b>utility/payoff values</b> to evaluate game outcomes.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |
| Agents involved       | Single decision-making agent.                                         | Two or more competing agents (e.g., MAX vs MIN players).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |        |                            |                               |                       |                                                                       |                                                                                           |                |                                                     |                                                                                    |                 |                                                              |                                                                                        |            |                                                  |                                                              |            |                                                    |                                                              |                 |                               |                                                          |         |  |

- **MAX player** → tries to maximize the score
- **MIN player** → tries to minimize the score

#### Working:

1. Expand the game tree fully.
2. Evaluate terminal nodes using utility function.
3. Propagate values upward.
  - MAX chooses maximum
  - MIN chooses minimum

Draw the Game Tree Example

#### Alpha-Beta Pruning (2 Marks)

Alpha-Beta pruning improves Minimax by cutting off branches that cannot affect the final decision.

- **Alpha ( $\alpha$ )**: Best value MAX can guarantee so far.
- **Beta ( $\beta$ )**: Best value MIN can guarantee so far.

#### Pruning Condition:

If

$$\beta \leq \alpha$$

Stop exploring that branch.

Alpha-Beta Pseudo-code: function AlphaBeta(node, depth,  $\alpha$ ,  $\beta$ , maximizingPlayer):

```

if depth = 0 or node is terminal:
 return utility(node)

if maximizingPlayer:
 value = $-\infty$
 for each child:
 value = max(value, AlphaBeta(child, depth-1, α , β , False))
 α = max(α , value)
 if $\beta \leq \alpha$:
 break # Beta cut-off
 return value

else:
 value = $+\infty$
 for each child:
 value = min(value, AlphaBeta(child, depth-1, α , β , True))
 β = min(β , value)
 if $\beta \leq \alpha$:
 break # Alpha cut-off
 return value

```

Minimax Algorithm – Pseudo-code

function MINIMAX(node, depth, maximizingPlayer):

```

if depth = 0 or node is terminal:
 return utility(node)

if maximizingPlayer:
 value = $-\infty$
 for each child of node:
 value = max(value, MINIMAX(child, depth-1, False))
 return value

else:
 value = $+\infty$
 for each child of node:

```

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |         |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|     | <pre>value = min(value, MINIMAX(child, depth-1, True)) return value</pre> <p><b>Minimax Algorithm</b></p> <p><b>Advantages:</b></p> <ul style="list-style-type: none"> <li>• Guarantees optimal move (if tree fully explored)</li> <li>• Works well for deterministic, zero-sum games</li> <li>• Simple and systematic approach</li> <li>• Provides clear decision-making strategy</li> </ul> <p><b>Limitations:</b></p> <ul style="list-style-type: none"> <li>• Very high time complexity: <math>O(bd)</math></li> <li>• Slow for large game trees</li> <li>• Requires complete tree exploration</li> <li>• Not suitable for real-time large games without optimization</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |         |
|     | <p><b>Alpha-Beta Pruning</b></p> <p><b>Advantages:</b></p> <ul style="list-style-type: none"> <li>• Produces same result as Minimax</li> <li>• Evaluates fewer nodes</li> <li>• Allows deeper search in same time</li> <li>• Improves efficiency significantly</li> <li>• Best performance with good move ordering</li> </ul> <p><b>Limitations:</b></p> <ul style="list-style-type: none"> <li>• Performance depends on move ordering</li> <li>• Worst case same as Minimax</li> <li>• More complex to implement</li> <li>• Still exponential time complexity</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |         |
|     | <p><b>(OR)</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
| (b) | <p>i) Explain the Backtracking algorithm for solving CSPs and discuss the importance of variable ordering and inference mechanisms. (7 Marks)</p> <p>ii) Explain how MRV and Forward Checking techniques improve the performance of CSP solvers with suitable examples. (6 Marks)</p> <p><b>(i) Backtracking Search Algorithm for a CSP (7 Marks)</b></p> <p><b>Algorithm Idea</b></p> <ul style="list-style-type: none"> <li>• Backtracking search is a <b>depth-first search</b> for CSPs.</li> <li>• It assigns values to variables one by one, checking consistency with constraints.</li> <li>• If a conflict occurs, it <b>backtracks</b> to try a different value</li> </ul> <p><b>Pseudo-code (Generic CSP)</b></p> <pre>function Backtracking-Search(csp):     return Backtrack({}, csp)  function Backtrack(assignment, csp):     if assignment is complete:         return assignment      var ← Select-Unassigned-Variable(csp, assignment) # Selection step     for each value in Order-Domain-Values(var, assignment, csp):         if value is consistent with assignment:             assignment[var] ← value             inferences ← Inference(csp, var, value) # Inference step</pre> | CO<br>3 |

```

if inferences ≠ failure:
 add inferences to assignment
 result ← Backtrack(assignment, csp)
 if result ≠ failure:
 return result
 remove var and inferences from assignment
return failure

```

### Role of Selection & Inference Steps

#### 1. Selection (Variable Selection):

- Chooses which variable to assign next.
- Smart selection reduces branching (e.g., MRV heuristic chooses the variable with the fewest legal values).

#### 2. Inference (Constraint Propagation):

- After assigning a variable, inference reduces the search space.
- Example: Forward Checking removes inconsistent values from neighbors' domains.
- Prevents exploring doomed partial assignments.

### (ii) MRV Heuristic & Forward Checking (6 Marks)

#### Minimum Remaining Values (MRV)

- Also called the **"fail-first" heuristic**.
- Selects the variable with the **fewest legal values left**.
- Motivation: Choose the most constrained variable first → if it fails, fail early and avoid wasted search.
- Example: In Sudoku, pick the empty cell with the least possible digits.

#### Forward Checking

- Each time a variable is assigned, eliminate values from neighboring variables that violate constraints.
- If any variable's domain becomes empty → backtrack immediately.
- Helps detect failure early before deep exploration.

#### Integration into Backtracking

- **MRV** guides the **selection step** (choosing the next variable smartly).
- **Forward Checking** enhances the **inference step** (constraint propagation after each assignment).
- Together, they:
  - Reduce search depth.
  - Cut off dead ends early.
  - Greatly improve efficiency compared to naive backtracking.

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| 14 | (a) | <p>i) Explain the architecture and components of a knowledge-based agent used in intelligent systems. (6 Marks)</p> <p>ii) Explain propositional theorem proving with an example and discuss its application in game-playing or adversarial decision-making systems. (7 Marks)</p> <p><b>Knowledge-Based Agent and Its Components (6 marks)</b></p> <p>A knowledge-based agent is an intelligent AI system that reasons, learns, and makes decisions using an explicit knowledge base combined with logical inference mechanisms. Its key power is the ability to store, update, and apply knowledge to solve complex problems rather than simply following fixed rules.</p> <p><b>Main Components</b></p> <ul style="list-style-type: none"> <li>• <b>Knowledge Base (KB):</b><br/>This is the central repository that holds structured information—facts, rules, and relationships—about the world or the agent's domain. The knowledge base is often represented using formal languages such as propositional or first-order logic and is updated as the agent learns or perceives new data from the environment.</li> <li>• <b>Inference Engine:</b><br/>The inference engine is the reasoning component that draws conclusions and makes decisions by manipulating and processing the information in the knowledge base. It applies logical operations, such as forward or backward chaining, to infer new knowledge and determine the best actions the agent should take.</li> <li>• <b>Perception and Action Modules (Sensors and Actuators):</b><br/>Sensors allow the agent to perceive or gather information from its environment, while actuators enable it to act or perform tasks in the environment based on its decisions. These modules ensure the agent can interact with the real world.</li> <li>• <b>Learning Module (optional):</b><br/>In some architectures, a learning module lets the agent acquire new knowledge or update its knowledge base from experiences or feedback, improving its performance over time.</li> </ul> <p><b>How Components Work Together</b></p> <ul style="list-style-type: none"> <li>• The agent uses its sensors to perceive its environment and collects information.</li> <li>• It retrieves relevant information from the knowledge base.</li> <li>• The inference engine reasons over this knowledge to derive new facts or make decisions.</li> <li>• The agent uses its actuators to implement the chosen actions in the environment.</li> <li>• New experiences or perceptions update the knowledge base, enhancing future reasoning.</li> </ul> <p>Knowledge-based agents are thus suitable for tasks that require flexible, context-aware problem-solving such as medical diagnosis, troubleshooting, and other expert-system domains.</p> <p>ii) Propositional Theorem Proving (7 marks)</p> <p>Propositional theorem proving is the process of determining whether a conclusion logically follows from a set of given propositional statements using formal rules of inference. It provides a systematic way to verify entailment in propositional logic.</p> <p><b>Process Steps</b></p> <ul style="list-style-type: none"> <li>• <b>Knowledge Representation:</b> All premises and the conclusion are expressed in symbolic form using propositional variables and logical operators (<math>\wedge</math>, <math>\vee</math>, <math>\neg</math>, <math>\rightarrow</math>).</li> </ul> | CO<br>4 |
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|  |          | <ul style="list-style-type: none"> <li>• <b>Conversion:</b> Sentences are often converted to a standardized form (such as Conjunctive Normal Form) to apply proof techniques such as resolution.</li> <li>• <b>Inference:</b> Apply valid inference rules (like Modus Ponens, Modus Tollens) or algorithms (like truth tables, resolution, or Wang's algorithm) to derive new sentences from the premises.</li> <li>• <b>Proof Construction:</b> Sequence these inferences to show how the conclusion follows from the premises, or to construct a contradiction from the negation of the conclusion.</li> <li>• <b>Termination:</b> If the conclusion is derived, the theorem is proved. If the process fails or produces inconsistency, the conclusion doesn't logically follow.</li> </ul> <p><b>Steps in Theorem Proving (Resolution Method):</b></p> <ol style="list-style-type: none"> <li>1. <b>Express premises and theorem in propositional logic.</b></li> <li>2. <b>Convert statements into Conjunctive Normal Form (CNF).</b></li> <li>3. <b>Negate the statement to be proved</b> and add it to the premises.</li> <li>4. <b>Apply resolution rule</b> repeatedly to derive contradictions.</li> <li>5. If a contradiction is found, the theorem is proved true.</li> </ol> <p><b>Example:</b><br/>Premises:</p> <ol style="list-style-type: none"> <li>1. <math>P \rightarrow Q</math></li> <li>2. <math>P</math></li> </ol> <p>Theorem to prove: <math>Q</math>.</p> <p><b>Step 1: Convert premises to CNF</b></p> <ul style="list-style-type: none"> <li>• <math>P \rightarrow Q</math> is equivalent to <math>\neg P \vee Q</math></li> <li>• <math>P</math> remains as it is.</li> </ul> <p><b>Step 2: Negate the theorem</b></p> <ul style="list-style-type: none"> <li>• Theorem: <math>Q</math>. Negation: <math>\neg Q</math>.</li> </ul> <p><b>Step 3: Apply resolution</b></p> <ul style="list-style-type: none"> <li>• From <math>\neg P \vee Q</math> and <math>\neg Q</math> resolve to get <math>\neg P</math></li> <li>• From <math>P</math> and <math>\neg P</math>, contradiction is reached.</li> </ul> <p>Therefore, <math>Q</math> is proven.</p> |         |
|  |          | <b>(OR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
|  | (b)<br>) | <p>i) Define the concepts of Syntax, Semantics, Entailment, and Model in Propositional Logic. (4 Marks)</p> <p>ii) Using truth tables, differentiate logical equivalence and logical entailment with examples, and explain their role in resolution-based theorem proving. (9 Marks)</p> <p><b>Syntax</b> defines the rules for forming valid sentences (well-formed formulas) in propositional logic.<br/>It specifies how symbols, propositional variables, and logical connectives (<math>\neg</math>, <math>\wedge</math>, <math>\vee</math>, <math>\rightarrow</math>, <math>\leftrightarrow</math>) can be combined.</p> <p><b>Semantics</b> defines the meaning of logical sentences.<br/>It assigns truth values (True/False) to propositions using interpretations or truth tables.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO<br>4 |

|    |     | <p>A <b>Model</b> is an assignment of truth values to propositional variables that makes a sentence true.</p> <p>If a sentence is true under a particular interpretation, that interpretation is called a model of the sentence.</p> <p><b>Entailment</b> means that a sentence logically follows from another.</p> <p>If every model of KB (Knowledge Base) also makes sentence <math>\alpha</math> true, then:<br/><math>KB \models \alpha</math></p> <p>It means <math>\alpha</math> is logically implied by KB.</p> <p><b>ii). Logical Equivalence (<math>\equiv</math>)</b></p> <p><b>Definition:</b></p> <p>Two propositions A and B are <b>logically equivalent</b> if they have the <b>same truth value in every possible model</b>.</p> <p>Formally: <math>A \equiv B \iff A \models B</math> and <math>B \models A</math></p> <p><math>\neg(P \vee Q) \equiv (\neg P \wedge \neg Q)</math>(De Morgan's Law)</p> <p><b>Truth Table:</b></p> <table><tr><th>P</th><th>Q</th><th><math>P \vee Q</math></th><th><math>\neg(P \vee Q)</math></th><th><math>\neg P</math></th><th><math>\neg Q</math></th><th><math>\neg P \wedge \neg Q</math></th></tr><tr><td>T</td><td>T</td><td>T</td><td>F</td><td>F</td><td>F</td><td>F</td></tr><tr><td>T</td><td>F</td><td>T</td><td>F</td><td>F</td><td>T</td><td>F</td></tr><tr><td>F</td><td>T</td><td>T</td><td>F</td><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>F</td><td>F</td><td>T</td><td>T</td><td>T</td><td>T</td></tr></table> <p>Columns for <math>\neg(P \vee Q)</math> and <math>\neg P \wedge \neg Q</math> are <b>identical</b> <math>\rightarrow</math> they are logically equivalent.</p> <p><b>2. Logical Entailment (<math>\models</math>)</b></p> <p><b>Definition:</b></p> <p>A proposition A <b>entails</b> proposition B (written <math>A \models B</math>) if <b>whenever A is true, B is also true</b>. This means <b>all models of A are also models of B</b>.</p> <p><b>Example:</b></p> <p><math>(P \wedge Q) \models P</math></p> <p><b>Truth Table:</b></p> <table><tr><th>P</th><th>Q</th><th><math>P \wedge Q</math></th><th>P</th></tr><tr><td>T</td><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td><td>T</td></tr><tr><td>F</td><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>F</td><td>F</td><td>F</td></tr></table> <p>In every row where <math>P \wedge Q</math> is <b>true</b> (first row), P is also <b>true</b>.</p> <p>Thus, <math>(P \wedge Q) \models P</math></p> <p><b>Logical Equivalence (<math>\equiv</math>):</b> Two formulas have <b>exactly the same truth values</b> in all rows of the truth table.</p> <p><b>Logical Entailment (<math>\models</math>):</b> When the first formula is true, the second must also be true (but not necessarily vice versa).</p> | P                | Q        | $P \vee Q$ | $\neg(P \vee Q)$       | $\neg P$ | $\neg Q$ | $\neg P \wedge \neg Q$ | T | T | T | F | F | F | F | T | F | T | F | F | T | F | F | T | T | F | T | F | F | F | F | F | T | T | T | T | P | Q | $P \wedge Q$ | P | T | T | T | T | T | F | F | T | F | T | F | F | F | F | F | F |  |
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| P  | Q   | $P \vee Q$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | $\neg(P \vee Q)$ | $\neg P$ | $\neg Q$   | $\neg P \wedge \neg Q$ |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| T  | T   | T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | F                | F        | F          | F                      |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| T  | F   | T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | F                | F        | T          | F                      |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| F  | T   | T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | F                | T        | F          | F                      |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| F  | F   | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | T                | T        | T          | T                      |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| P  | Q   | $P \wedge Q$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | P                |          |            |                        |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| T  | T   | T                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | T                |          |            |                        |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| T  | F   | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | T                |          |            |                        |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| F  | T   | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | F                |          |            |                        |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| F  | F   | F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | F                |          |            |                        |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| 15 | (a) | <p>i) Explain the concept of classical planning and describe its importance in AI. (5 Marks)</p> <p>ii) Illustrate Partial Order Planning with an example, explaining its advantages over total-order planning. (8 Marks)</p> <p>Classical planning is a structured approach in AI where an agent formulates a sequence of actions to achieve a specific goal from a given initial state. It assumes a fully observable, deterministic, and static environment, where the effects of actions are known in advance.</p> <p>Key Features of Classical Planning:</p> <ul style="list-style-type: none"><li>• States: Representations of the world at a given time.</li><li>• Actions: Defined by preconditions and effects.</li><li>• Goals: Desired conditions to be achieved.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO<br>5          |          |            |                        |          |          |                        |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |              |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |

- Plans: Ordered sequences of actions leading from the initial state to the goal.

## 2. Importance of Classical Planning in AI

Efficient Problem Solving: AI can automate complex decision-making tasks.

Predictability: Since the environment is deterministic, the outcomes of actions are known.

Applicability: Used in robotics, logistics, automated scheduling, and game AI.

Example of Classical Planning:

A robot in a warehouse needs to pick up a package and place it in a delivery zone.

- Initial state: The package is at location A.
- Goal state: The package is at location B.
- Actions: Move(A → B), PickUp(package), Drop(package).
- Plan: PickUp(package) → Move(A → B) → Drop(package).

## (ii) Partial Order Planning with Example

### 1. What is Partial Order Planning?

Partial Order Planning (POP) is an AI planning approach where actions are only ordered when necessary, allowing flexibility in execution. Unlike Total Order Planning, where every action is strictly sequenced, POP ensures that only dependent actions are ordered while independent actions remain flexible.

Key Features of Partial Order Planning:

- Only necessary constraints are imposed between actions.
- Parallel execution is possible for independent actions.
- Uses causal links to ensure preconditions are met before an action is executed.

### 2. Example of Partial Order Planning

Scenario: Making Breakfast

Goal: Make toast and coffee.

Actions & Constraints:

- Toast Bread (T) → Requires Bread in Toaster.
- Brew Coffee (C) → Requires Water and Coffee.
- Spread Butter (B) → Requires Toasted Bread.
- Serve Breakfast (S) → Requires Coffee & Buttered Toast.

Partial Order Plan:

- "Toast Bread" and "Brew Coffee" can be done in parallel.
- "Spread Butter" must follow "Toast Bread".
- "Serve Breakfast" must wait for all tasks to be completed.

Advantages of Partial Order Planning over Total Order Planning:

More flexible execution – Allows parallel processing.

Better efficiency – Reduces unnecessary ordering constraints.

Handles complex problems – Works well in multi-agent and real-time AI planning scenarios.

### 3. Total Order vs. Partial Order Planning Comparison

| Feature         | Total Order Planning            | Partial Order Planning             |
|-----------------|---------------------------------|------------------------------------|
| Execution Order | Strict sequence                 | Only necessary ordering            |
| Parallelism     | No parallel execution           | Allows parallel execution          |
| Efficiency      | Less efficient in complex tasks | More scalable and flexible         |
| Example         | Step 1 → Step 2 → Step 3        | Step 1, Step 2 (parallel) → Step 3 |

(OR)

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
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| (b)<br>) | <p>i) Explain STRIPS (Stanford Research Institute Problem Solver), its representation, and how it is used in AI planning. (6 Marks)</p> <p>ii) Discuss planning and acting in non-deterministic domains, highlighting the challenges and solutions with an example. (7 Marks)</p> <p>STRIPS (Stanford Research Institute Problem Solver) is a formal language used in AI for automated planning. It provides a structured way to represent states, goals, and actions, enabling AI systems to generate sequences of actions (plans) to achieve a desired outcome.</p> <p><b>Key Features of STRIPS:</b></p> <ul style="list-style-type: none"> <li>• Uses predicate logic to define states and actions.</li> <li>• Represents planning problems in terms of initial state, goal state, and operators (actions).</li> <li>• Provides an efficient way to model complex environments.</li> </ul> <p><b>2. Representation in STRIPS</b><br/> A STRIPS planning problem consists of:<br/> Initial State (I) → Describes the world at the start.<br/> Goal State (G) → Specifies conditions that must be met.<br/> Operators (O) → Defines possible actions using:</p> <ul style="list-style-type: none"> <li>• Preconditions (What must be true before the action)</li> <li>• Add List (Facts added after the action)</li> <li>• Delete List (Facts removed after the action)</li> </ul> <p><b>3. Example: Robot Moving an Object</b><br/> <b>Problem:</b> A robot moves a box from Room A to Room B<br/> <b>STRIPS Representation:</b></p> <ul style="list-style-type: none"> <li>• Initial State: <math>At(Robot, RoomA) \wedge At(Box, RoomA)</math></li> <li>• Goal State: <math>At(Box, RoomB)</math></li> <li>• Action: <math>Move(Robot, RoomA, RoomB)</math> <ul style="list-style-type: none"> <li>◦ Precondition: <math>At(Robot, RoomA)</math></li> <li>◦ Add List: <math>At(Robot, RoomB)</math></li> <li>◦ Delete List: <math>At(Robot, RoomA)</math></li> </ul> </li> <li>• Action: <math>Push(Box, RoomA, RoomB)</math> <ul style="list-style-type: none"> <li>◦ Precondition: <math>At(Robot, RoomA) \wedge At(Box, RoomA)</math></li> <li>◦ Add List: <math>At(Box, RoomB)</math></li> <li>◦ Delete List: <math>At(Box, RoomA)</math></li> </ul> </li> </ul> <p><b>How STRIPS Works Here:</b><br/> The AI planner checks the goal (<math>At(Box, RoomB)</math>).<br/> It identifies a valid action (<math>Push(Box, RoomA, RoomB)</math>).<br/> Since Push requires <math>At(Robot, RoomA)</math>, it ensures the robot moves first.<br/> The final plan is:</p> <ul style="list-style-type: none"> <li>• <math>Move(Robot, RoomA, RoomB)</math></li> <li>• <math>Push(Box, RoomA, RoomB)</math></li> </ul> <p>STRIPS allows AI to generate structured, logical plans efficiently!</p> <p>(ii) Planning and Acting in Non-Deterministic Domains (7 Marks)</p> <p><b>1. What Are Non-Deterministic Domains?</b><br/> A non-deterministic domain is an environment where actions may have multiple possible outcomes, making planning more complex.</p> <p><b>Challenges in Non-Deterministic Domains:</b><br/> Uncertainty – The same action may lead to different results.<br/> Multiple possible futures – AI must plan for various outcomes.<br/> Dynamic adaptation – AI must adjust when unexpected changes occur.</p> | CO<br>5 |
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|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |  |
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|  | <p>2. Solutions for Non-Deterministic Planning</p> <p>Conditional Planning – AI creates multiple branches for different outcomes.</p> <p>Execution Monitoring – AI checks the results of actions and adapts.</p> <p>Replanning – If the unexpected happens, AI modifies its plan dynamically.</p> <p>Probabilistic Models – AI assigns probabilities to different outcomes.</p> <p>3. Example: A Robot Navigating a Slippery Floor</p> <p>Problem:</p> <p>A robot wants to move from Start to Goal, but some tiles are slippery, making movement uncertain.</p> <p>Actions &amp; Possible Outcomes:</p> <p>Move(Forward) → Success (80%)   Slip Back (20%)</p> <p>Move(Left) → Success (90%)   Slip (10%)</p> <p>Solution Using Non-Deterministic Planning:</p> <p>The AI creates a conditional plan:</p> <ul style="list-style-type: none"> <li>• If Move(Forward) succeeds → Continue.</li> <li>• If it slips back, then try a different route.</li> </ul> <p>AI monitors the result and adjusts its plan dynamically.</p> <p>Probabilities help the AI choose the safest path.</p> <p>By handling multiple possible outcomes, AI can make robust decisions in uncertain environments</p> |  |
|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

**PART C** **(1 x 15 = 15 marks)**

**(Case study/Comprehensive type Questions)**

|       |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>CO</b> |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
|-------|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|------|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|------------|
| 16.   | (a)  | <p>Suppose you have the following search space:</p> <p>a) Draw the state space for this problem.</p> <p>b) Assume that the Initial state is A and the goal state is G. Show how various Best First search algorithms would create a search tree to find a path from the initial state to reach the goal state:</p> <table><tr><th>STATE</th><th>NEXT</th><th>COST</th></tr><tr><td>A</td><td>B</td><td>4</td></tr><tr><td>A</td><td>C</td><td>1</td></tr><tr><td>B</td><td>D</td><td>3</td></tr><tr><td>B</td><td>E</td><td>8</td></tr><tr><td>C</td><td>C</td><td>0</td></tr><tr><td>C</td><td>D</td><td>2</td></tr><tr><td>C</td><td>F</td><td>6</td></tr><tr><td>D</td><td>C</td><td>2</td></tr><tr><td>D</td><td>E</td><td>4</td></tr><tr><td>E</td><td>G</td><td>2</td></tr></table> | STATE     | NEXT | COST | A | B | 4 | A | C | 1 | B | D | 3 | B | E | 8 | C | C | 0 | C | D | 2 | C | F | 6 | D | C | 2 | D | E | 4 | E | G | 2 | <b>CO2</b> |
| STATE | NEXT | COST                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| A     | B    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| A     | C    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| B     | D    | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| B     | E    | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| C     | C    | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| C     | D    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| C     | F    | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| D     | C    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| D     | E    | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |
| E     | G    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |      |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |            |

|       |      | <table><tr><td>F</td><td>G</td><td>8</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | F     | G    | 8 |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|-------|------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|------|---|---|---|---|---|---|---|---|---|---|---|---|---|---|--|
| F     | G    | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|       |      | <p><b>Heuristics:</b></p> <table><tr><th>STATE</th><th>H(N)</th></tr><tr><td>A</td><td>8</td></tr><tr><td>B</td><td>8</td></tr><tr><td>C</td><td>6</td></tr><tr><td>D</td><td>5</td></tr><tr><td>E</td><td>1</td></tr><tr><td>F</td><td>4</td></tr><tr><td>G</td><td>0</td></tr></table> <p>Answer:</p> <p>i) Uniform Cost Search: ACDEG</p> <p>ii) Greedy Search: ACFG</p> <p>iii) A* Search: ACDEG</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | STATE | H(N) | A | 8 | B | 8 | C | 6 | D | 5 | E | 1 | F | 4 | G | 0 |  |
| STATE | H(N) |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| A     | 8    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| B     | 8    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| C     | 6    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| D     | 5    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| E     | 1    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| F     | 4    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
| G     | 0    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|       |      | <p style="text-align: center;"><b>(OR)</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |
|       | (b)  | <p>i) What are local search and optimization problems? Explain how local search operates in continuous state spaces. (8 Marks)</p> <p>ii) Compare Simulated Annealing and Genetic Algorithms in terms of exploration, exploitation, and convergence behavior. (7 Marks)</p> <p><b>Local Search (2 Marks)</b></p> <p>Local search algorithms:</p> <ul style="list-style-type: none"><li>Start with an initial solution.</li><li>Move to a neighboring solution.</li><li>Try to improve the objective (evaluation) function.</li></ul> <p>They do not construct paths but directly search in the state space.</p> <p>Examples: Hill Climbing, Simulated Annealing.</p> <p><b>Optimization Problems (2 Marks)</b></p> <p>An optimization problem aims to:</p> <ul style="list-style-type: none"><li>Find the <b>best solution</b> (maximum or minimum value).</li><li>Based on an objective function.</li></ul> <p>Examples:</p> <ul style="list-style-type: none"><li>Traveling Salesman Problem</li><li>Job scheduling</li><li>Machine learning parameter tuning</li></ul> <p><b>Working of Local Search (2 Marks)</b></p> <ol style="list-style-type: none"><li>Start from an initial state.</li><li>Evaluate neighboring states.</li><li>Move to a better neighbor.</li><li>Repeat until no improvement is possible.</li></ol> <p>It may get stuck in:</p> <ul style="list-style-type: none"><li>Local maxima</li><li>Plateaus</li><li>Ridges</li></ul> <p><b>Local Search in Continuous State Spaces (2 Marks)</b></p> <p>In continuous problems:</p> <ul style="list-style-type: none"><li>States are real-valued (not discrete).</li><li>The algorithm uses <b>gradient information</b>.</li></ul> | CO2   |      |   |   |   |   |   |   |   |   |   |   |   |   |   |   |  |

|                     | <p>Example:<br/>Gradient Descent</p> <p>Steps:</p> <ul style="list-style-type: none"> <li>• Compute slope (gradient).</li> <li>• Move in direction of improvement.</li> <li>• Continue until convergence.</li> </ul> <p>Used in:</p> <ul style="list-style-type: none"> <li>• Neural networks</li> <li>• Function optimization</li> </ul> <p>(ii) Simulated Annealing vs Genetic Algorithms (7 Marks)</p> <p><b>Simulated Annealing (2 Marks)</b></p> <p>Simulated Annealing is a probabilistic local search algorithm inspired by the metal cooling process.</p> <ul style="list-style-type: none"> <li>• Starts with a high temperature (T).</li> <li>• Accepts worse solutions with probability:</li> </ul> $P = e^{-\Delta E/T}$ <ul style="list-style-type: none"> <li>• As temperature decreases, it becomes greedy.</li> <li>• Helps escape local optima.</li> </ul> <p>It works on a <b>single solution</b> at a time.</p> <p><b>Genetic Algorithms (2 Marks)</b></p> <p>Genetic Algorithm (GA) is a population-based search technique inspired by natural evolution.</p> <p>Main operators:</p> <ul style="list-style-type: none"> <li>• Selection</li> <li>• Crossover</li> <li>• Mutation</li> </ul> <p>It maintains multiple candidate solutions and evolves them over generations.</p> <p><b>Comparison (3 Marks)</b></p> <table> <tr> <th>Feature</th><th>Simulated Annealing</th><th>Genetic Algorithm</th></tr> <tr> <td>Search Type</td><td>Single solution</td><td>Population-based</td></tr> <tr> <td>Exploration</td><td>Controlled by temperature</td><td>High due to population diversity</td></tr> <tr> <td>Exploitation</td><td>Gradual improvement of one solution</td><td>Combines best individuals</td></tr> <tr> <td>Escape Local Optima</td><td>Yes (probabilistic moves)</td><td>Yes (mutation &amp; crossover)</td></tr> <tr> <td>Convergence</td><td>Slow and steady</td><td>Faster but may converge prematurely</td></tr> <tr> <td>Memory Usage</td><td>Low</td><td>Higher</td></tr> </table> | Feature                             | Simulated Annealing | Genetic Algorithm | Search Type | Single solution | Population-based | Exploration | Controlled by temperature | High due to population diversity | Exploitation | Gradual improvement of one solution | Combines best individuals | Escape Local Optima | Yes (probabilistic moves) | Yes (mutation & crossover) | Convergence | Slow and steady | Faster but may converge prematurely | Memory Usage | Low | Higher |  |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|---------------------|-------------------|-------------|-----------------|------------------|-------------|---------------------------|----------------------------------|--------------|-------------------------------------|---------------------------|---------------------|---------------------------|----------------------------|-------------|-----------------|-------------------------------------|--------------|-----|--------|--|
| Feature             | Simulated Annealing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Genetic Algorithm                   |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |
| Search Type         | Single solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Population-based                    |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |
| Exploration         | Controlled by temperature                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | High due to population diversity    |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |
| Exploitation        | Gradual improvement of one solution                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Combines best individuals           |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |
| Escape Local Optima | Yes (probabilistic moves)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Yes (mutation & crossover)          |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |
| Convergence         | Slow and steady                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Faster but may converge prematurely |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |
| Memory Usage        | Low                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Higher                              |                     |                   |             |                 |                  |             |                           |                                  |              |                                     |                           |                     |                           |                            |             |                 |                                     |              |     |        |  |

|                  | For Set 2 AK |       |       |       |       |       |       |       |       |       |          |       |
|------------------|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|-------|
|                  | Part - A     |       |       |       |       |       |       |       |       |       |          |       |
| Question No.     | 1            | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |          |       |
| Knowledge Level  | K1           | K2    | K1    | K1    | K1    | K2    | K2    | K1    | K1    | K1    |          |       |
| Difficulty Level | 1            | 2     | 1     | 1     | 1     | 2     | 1     | 1     | 1     | 1     |          |       |
|                  | Part - B     |       |       |       |       |       |       |       |       |       | Part - C |       |
| Question No.     | 11(a)        | 11(b) | 12(a) | 12(b) | 13(a) | 13(b) | 14(a) | 14(b) | 15(a) | 15(b) | 16(a)    | 16(b) |
| Knowledge Level  | K5           | K3    | K4    | K4    | K4    | K4    | K4    | K4    | K4    | K4    | K3       | K4    |
| Difficulty Level | 4            | 3     | 3     | 3     | 3     | 4     | 4     | 4     | 4     | 4     | 3        | 4     |

**SET 3**

**QP Setter Name / Dept:**

QP Code:

Reg. No

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**SAVEETHA ENGINEERING COLLEGE**

(Autonomous, Affiliated to Anna University, Chennai)

**B.E./B.Tech. End Semester Examinations – Apr/May 2026 (R2019/R2024)**

**Common to AI/ML, AI/DS, CSE, IT (Branches)**

**19AI405/CS3405-FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE**

Time: Three hours

Maximum marks: 100



**PART A**

**(10 x 2 = 20 marks)**

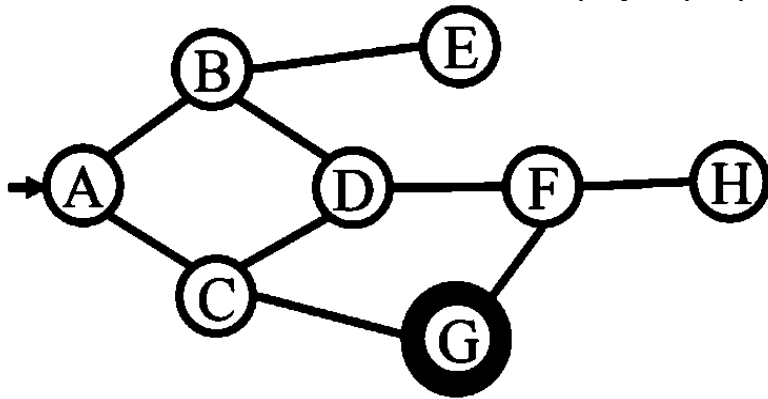
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|    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>CO</b>  |
| 1. | <p>Define Artificial Intelligence?<br/> “Artificial Intelligence is the science and engineering of making intelligent machines, especially intelligent computer programs.”</p> <ul style="list-style-type: none"> <li>It emphasizes both the scientific aspect (understanding intelligence) and the engineering aspect (building systems that act intelligently).</li> <li>This definition is widely accepted in AI textbooks and exams.</li> </ul> <p>AI = science + engineering of intelligent machines.</p>                                                                                                                                                                                                                                                                                                                                                              | <b>CO1</b> |
| 2. | <p>State the purpose of problem formulation in AI.<br/> <b>Problem formulation in AI is the process of clearly defining a problem in terms of:</b></p> <ul style="list-style-type: none"> <li><b>Initial state</b></li> <li><b>Actions</b></li> <li><b>Goal state</b></li> <li><b>Path cost</b></li> </ul> <p><b>The purpose of problem formulation is to:</b></p> <ul style="list-style-type: none"> <li><b>Convert a real-world problem into a structured search problem.</b></li> <li><b>Enable the agent to apply appropriate search algorithms to find a solution efficiently.</b></li> </ul>                                                                                                                                                                                                                                                                          | <b>CO1</b> |
| 3. | <p>Define optimization problem in AI.<br/> <b>An optimization problem in AI is a problem where the objective is to find the best possible solution among many possible solutions based on an objective (evaluation) function.</b><br/> <b>It aims to either:</b></p> <ul style="list-style-type: none"> <li><b>Maximize a function (e.g., profit, accuracy), or</b></li> <li><b>Minimize a function (e.g., cost, error).</b></li> </ul> <p><b>Example:</b></p> <ul style="list-style-type: none"> <li><b>Traveling Salesman Problem</b></li> <li><b>Machine learning model parameter tuning</b></li> </ul>                                                                                                                                                                                                                                                                  | <b>CO2</b> |
| 4. | <p>What is meant by heuristic search space? Illustrate with an example.<br/> It uses additional knowledge (heuristics) about the problem to guide the search toward the goal more efficiently, reducing the number of states explored.</p> <ul style="list-style-type: none"> <li>Leads to faster search and often finds the solution using less time and memory.</li> <li>Example: A* uses a heuristic to estimate the distance to the goal, unlike BFS or DFS, which explore blindly.</li> </ul> <p>Heuristic search is more efficient and goal-directed than uninformed search</p>                                                                                                                                                                                                                                                                                       | <b>CO2</b> |
| 5. | <p>Define a Constraint Satisfaction Problem (CSP) and list its three key components.<br/> A CSP is a type of problem where the goal is to assign values to a set of variables such that all constraints between the variables are satisfied.</p> <ul style="list-style-type: none"> <li>Common in AI for scheduling, map coloring, puzzles, and planning.</li> </ul> <p><b>Three Key Components of a CSP</b></p> <ol style="list-style-type: none"> <li><b>Variables (X):</b> The set of items to be assigned values (e.g., <math>X = \{A, B, C\}</math>).</li> <li><b>Domains (D):</b> The set of possible values for each variable (e.g., <math>A \in \{1,2,3\}</math>).</li> <li><b>Constraints (C):</b> Rules that restrict the values that variables can take simultaneously (e.g., <math>A \neq B</math>).</li> </ol> <p>CSP = Variables + Domains + Constraints.</p> | <b>CO3</b> |

| 6.                    | <p>Define Monte Carlo Tree Search (MCTS).</p> <p><b>Monte Carlo Tree Search (MCTS) is a heuristic search algorithm used in decision-making and game playing that builds a search tree using random simulations to estimate the best move.</b></p> <p><b>It uses repeated random sampling to evaluate the possible outcomes of actions and selects moves based on statistical results.</b></p> <p><b>Main Steps:</b></p> <ol style="list-style-type: none"> <li>1. Selection</li> <li>2. Expansion</li> <li>3. Simulation (Random Play)</li> <li>4. Backpropagation</li> </ol> <p><b>Used in games like Go and Chess.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO3                                                                                                                                                |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
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| 7.                    | <p>Define universal and existential quantifiers.</p> <p><b>1. Universal Quantifier (<math>\forall</math>)</b></p> <ul style="list-style-type: none"> <li>• Symbol: <math>\forall</math></li> <li>• Meaning: “for all” or “every”</li> <li>• Expresses that a <b>property or statement is true for all elements</b> in a domain.</li> <li>• <b>Example:</b><br/> <math>\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))</math></li> </ul> <p><b>2. Existential Quantifier (<math>\exists</math>)</b></p> <ul style="list-style-type: none"> <li>• Symbol: <math>\exists</math></li> <li>• Meaning: “there exists” or “for some”</li> <li>• Expresses that a <b>property or statement is true for at least one element</b> in a domain.</li> <li>• <b>Example:</b><br/> <math>\exists x (\text{Human}(x) \wedge \text{Intelligent}(x))</math></li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO4                                                                                                                                                |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| 8.                    | <p>State the difference between propositional logic and first-order logic.</p> <table border="1"> <thead> <tr> <th>Aspect</th><th>Propositional Logic (PL)</th><th>First-Order Logic (FOL)</th></tr> </thead> <tbody> <tr> <td><b>Expressiveness</b></td><td>Deals with <b>atomic propositions</b> (true/false statements).</td><td>Deals with <b>objects, relations, and quantifiers</b>, allowing richer statements.</td></tr> <tr> <td><b>Components</b></td><td>Propositions, logical connectives (<math>\wedge, \vee, \neg, \rightarrow</math>).</td><td>Variables, constants, functions, predicates, quantifiers (<math>\forall, \exists</math>), and logical connectives.</td></tr> <tr> <td><b>Quantifiers</b></td><td>Not available.</td><td>Supports <b>universal (<math>\forall</math>) and existential (<math>\exists</math>) quantifiers</b>.</td></tr> <tr> <td><b>Example</b></td><td><math>P \rightarrow Q \rightarrow Q</math> ("If it rains, the ground is wet")</td><td><math>\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))</math> for all x<br/> <math>(\text{Human}(x) \rightarrow \text{Mortal}(x))</math> ("All humans are mortal")</td></tr> <tr> <td><b>Applications</b></td><td>Simple decision problems, circuits, truth assignments.</td><td>Knowledge representation, AI reasoning, natural language understanding.</td></tr> </tbody> </table> | Aspect                                                                                                                                             | Propositional Logic (PL) | First-Order Logic (FOL) | <b>Expressiveness</b> | Deals with <b>atomic propositions</b> (true/false statements). | Deals with <b>objects, relations, and quantifiers</b> , allowing richer statements. | <b>Components</b> | Propositions, logical connectives ( $\wedge, \vee, \neg, \rightarrow$ ). | Variables, constants, functions, predicates, quantifiers ( $\forall, \exists$ ), and logical connectives. | <b>Quantifiers</b> | Not available. | Supports <b>universal (<math>\forall</math>) and existential (<math>\exists</math>) quantifiers</b> . | <b>Example</b> | $P \rightarrow Q \rightarrow Q$ ("If it rains, the ground is wet") | $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ for all x<br>$(\text{Human}(x) \rightarrow \text{Mortal}(x))$ ("All humans are mortal") | <b>Applications</b> | Simple decision problems, circuits, truth assignments. | Knowledge representation, AI reasoning, natural language understanding. | CO4 |
| Aspect                | Propositional Logic (PL)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | First-Order Logic (FOL)                                                                                                                            |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| <b>Expressiveness</b> | Deals with <b>atomic propositions</b> (true/false statements).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Deals with <b>objects, relations, and quantifiers</b> , allowing richer statements.                                                                |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| <b>Components</b>     | Propositions, logical connectives ( $\wedge, \vee, \neg, \rightarrow$ ).                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Variables, constants, functions, predicates, quantifiers ( $\forall, \exists$ ), and logical connectives.                                          |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| <b>Quantifiers</b>    | Not available.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Supports <b>universal (<math>\forall</math>) and existential (<math>\exists</math>) quantifiers</b> .                                              |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| <b>Example</b>        | $P \rightarrow Q \rightarrow Q$ ("If it rains, the ground is wet")                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ for all x<br>$(\text{Human}(x) \rightarrow \text{Mortal}(x))$ ("All humans are mortal") |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| <b>Applications</b>   | Simple decision problems, circuits, truth assignments.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Knowledge representation, AI reasoning, natural language understanding.                                                                            |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| 9.                    | <p>What is hierarchical planning?</p> <p><b>Hierarchical Planning is a planning approach in AI where a complex problem is broken down into smaller, manageable sub-tasks arranged in a hierarchy.</b></p> <ul style="list-style-type: none"> <li>• High-level tasks are decomposed into lower-level actions.</li> <li>• Planning is done step by step from abstract goals to concrete actions.</li> </ul> <p><b>Advantage:</b><br/> <b>Reduces complexity and makes planning more efficient.</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | CO5                                                                                                                                                |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |
| 10.                   | What is action schema in planning?                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | CO5                                                                                                                                                |                          |                         |                       |                                                                |                                                                                     |                   |                                                                          |                                                                                                           |                    |                |                                                                                                       |                |                                                                    |                                                                                                                                                    |                     |                                                        |                                                                         |     |

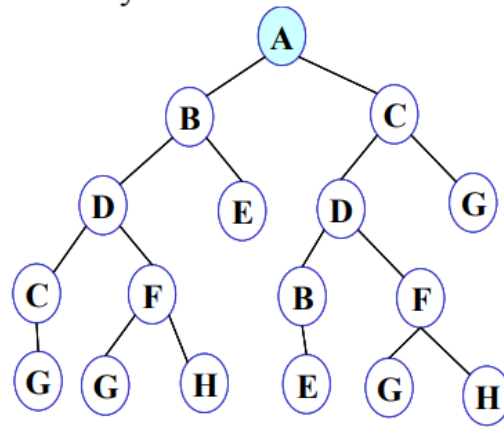
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|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p><b>An Action Schema in planning is a generalized representation of an action that describes:</b></p> <ul style="list-style-type: none"> <li>• <b>Action name</b></li> <li>• <b>Parameters (variables)</b></li> <li>• <b>Preconditions (conditions that must be true before execution)</b></li> <li>• <b>Effects (changes produced after execution)</b></li> </ul> <p><b>It allows one action to represent many specific actions using variables.</b></p> <p><b>Example</b></p> <p><b>Action:</b> Move(x, y)</p> <p><b>Precondition:</b> At(x) <math>\wedge</math> Connected(x,y)</p> <p><b>Effect:</b> <math>\neg</math>At(x) <math>\wedge</math> At(y)</p> <p><b>Used in classical planning to generate specific actions from general templates.</b></p> |  |
|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

## PART B

(5 x 13 = 65 marks)

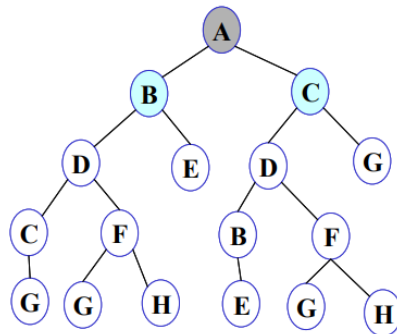
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|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>CO</b>  |
| 11. | (a) | <p>Explain the Depth-First Search (DFS) algorithm in detail. How does it function as an uninformed search strategy? Apply DFS to traverse the given state space graph with 'A' as Start Node and 'G' as Goal Node, and show the step-by-step expansion of nodes.</p>  <p>Depth-First Search (DFS) is an uninformed search algorithm that explores a search tree by expanding the deepest node first before backtracking. Unlike Breadth-First Search (BFS), which explores level by level, DFS follows a depth-first approach, making it more memory-efficient but less optimal in certain cases.</p> <p><b>Characteristics of DFS</b></p> <p>Complete: DFS is not always complete in infinite search spaces (it may get stuck in loops).</p> <p>Optimal: DFS is not optimal as it does not necessarily find the shortest path.</p> <p>Time Complexity: <math>O(b^d)</math> (where <math>b</math> = branching factor, <math>d</math> = depth of solution).</p> <p>Space Complexity: <math>O(d)</math> (only keeps track of nodes along the current path).</p> <p><b>Algorithm of DFS</b></p> <p><b>Steps of DFS:</b></p> <ol style="list-style-type: none"> <li>1. Start with the initial node and push it onto the stack.</li> <li>2. Expand the top node and push its child nodes onto the stack.</li> <li>3. Continue expanding the deepest (last added) node first.</li> <li>4. If a dead end is reached, backtrack to the previous node.</li> <li>5. Repeat until the goal node is found or the stack is empty (failure).</li> </ol> | <b>CO1</b> |

Step 1: Initially fringe contains only the node for A.



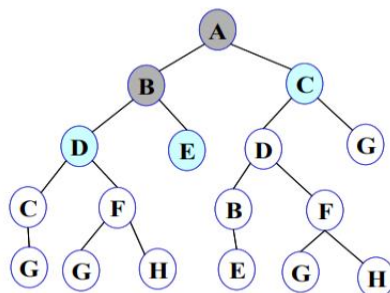
FRINGE: A

Step 2: A is removed from fringe. A is expanded and its children B and C are put in front of fringe.

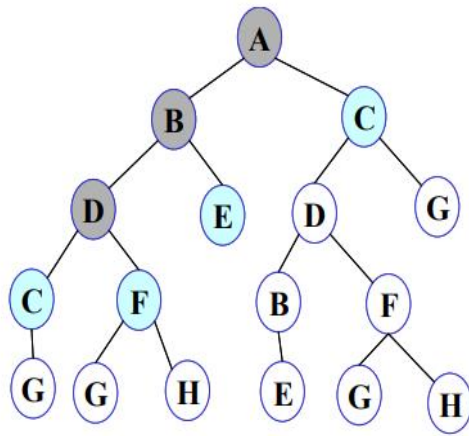


FRINGE: B C

Step 3: Node B is removed from fringe, and its children D and E are pushed in front of fringe.

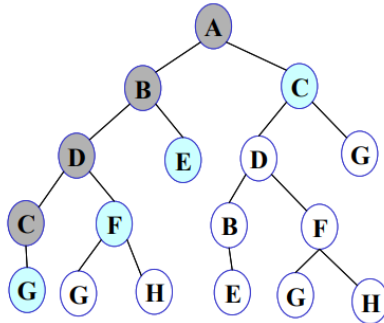


FRINGE: D E C



FRINGE: C F E C

Step 5: Node C is removed from fringe. Its child G is pushed in front of fringe.



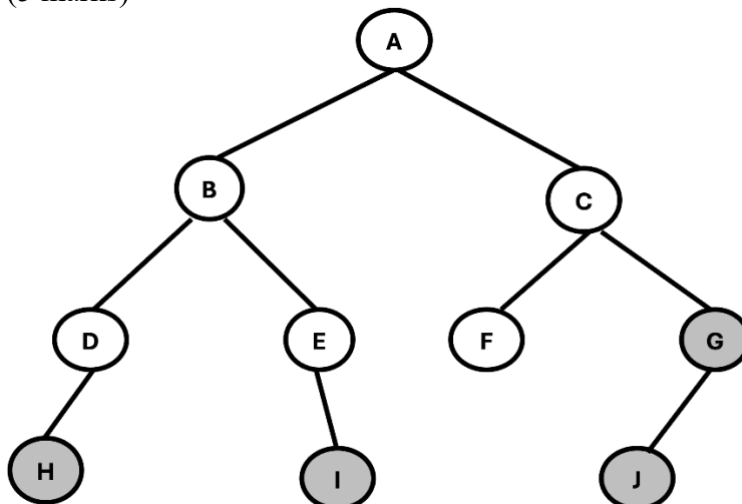
FRINGE: G F E C

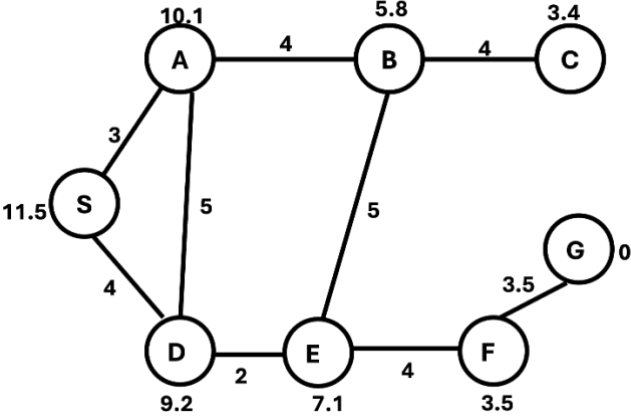
PATH- ABDCG

(OR)

(b) Describe the core principles of Depth-Limited Search (DLS) and Iterative Deepening Depth-First Search (IDDFS). Provide the standard pseudocode for the DLS recursive function, clearly explaining the role of its parameters. Furthermore, explain the key problem that IDDFS solves compared to a standard Depth-First Search.

- a) The state space graph below has A as its start node and G as the goal node. The connections and costs between nodes are defined in the associated diagram (5 marks)



|     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
|-----|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|     |     | <p>b) Apply the Iterative Deepening Search algorithm to this graph:</p> <ol style="list-style-type: none"> <li>Show the sequence of node expansions for each iterative depth level (beginning at depth 0) until the goal node <b>G</b> is found.</li> <li>For each depth level, state the limit and list the order of nodes visited.</li> <li>Report the final solution path from A to G discovered by IDS.</li> </ol> <p>(8 marks)</p> <p>DLS PATH:A-&gt;B-&gt;D-&gt;E-&gt;C-&gt;F-&gt;G<br/> IDDFS PATH:A-&gt;B-&gt;D-&gt;E-&gt;C-&gt;F-&gt;G</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |     |
| 12. | (a) | <p>i) Explain the A* search algorithm and write the pseudo code. (7 Marks)</p> <p>ii) Apply the A* algorithm to find the optimal path in the given State Space Graph (S → G). (6 Marks)</p>  <p>Answer:<br/> Path: SDEFG</p> <p>A* (A-Star) search is a best-first search algorithm that finds the shortest path from a start node to a goal node while ensuring optimality. It balances exploration (searching for better paths) and exploitation (following the best known path).</p> <p>Evaluation Function:<br/> A* uses a function <math>f(n)</math> to determine the next node to expand:<br/> <math>f(n) = g(n) + h(n)</math><br/> Where:</p> <ul style="list-style-type: none"> <li><math>g(n)</math> = Cost from the start node to the current node n (known cost).</li> <li><math>h(n)</math> = Estimated cost from n to the goal (heuristic).</li> </ul> <p>How It Works:</p> <ol style="list-style-type: none"> <li>Start from the initial node and compute <math>f(n)</math> for its neighbors.</li> <li>Select the node with the lowest <math>f(n)</math> value.</li> <li>Repeat until the goal node is reached.</li> </ol> | CO2 |

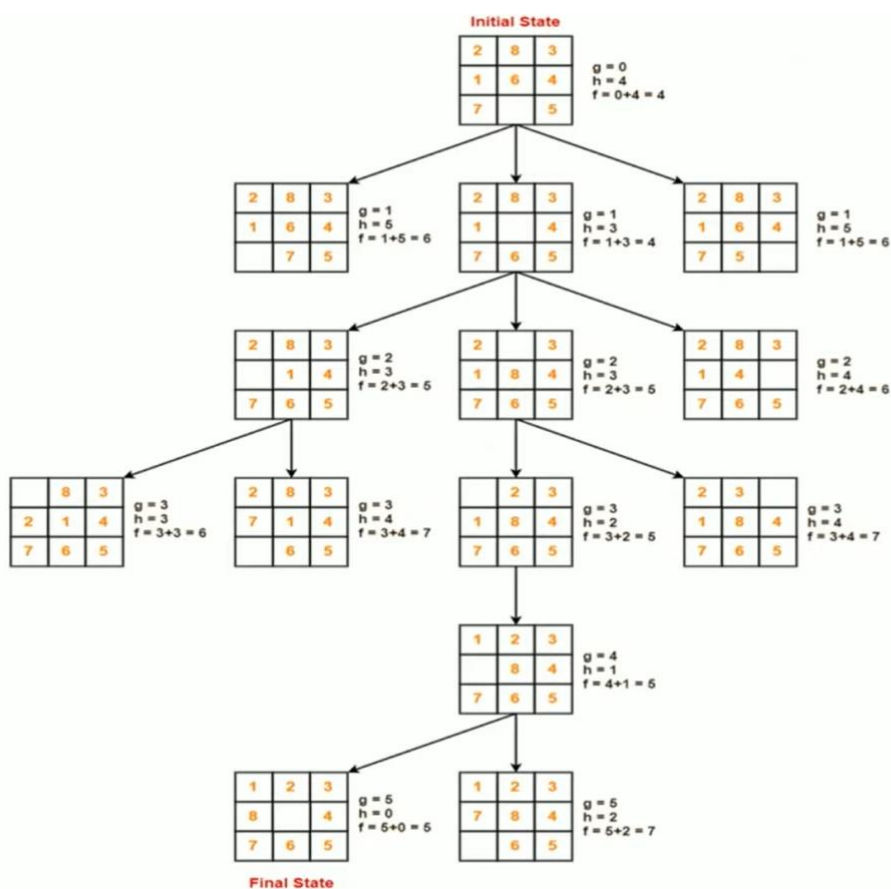
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|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|---|---|---|--|---|-----|
|     | <p>4. The algorithm guarantees the shortest path if <math>h(n)</math> is admissible (never overestimates the actual cost).</p> <p><i>Advantages of A Search:*</i><br/>Guaranteed Optimality (if heuristic is admissible and consistent).<br/>Efficient for many real-world problems (pathfinding, game AI, robotics).<br/>Avoids unnecessary expansions compared to Uniform Cost Search.</p> <p><i>Disadvantages of A Search:*</i><br/>Memory-Intensive – Stores all visited nodes, leading to high space complexity.<br/>Performance Depends on the Heuristic – If <math>h(n)</math> is poorly chosen, performance degrades.</p> <p>function A-STAR(start, goal):</p> <pre>open_set ← {start} came_from ← empty map  g(start) ← 0 f(start) ← g(start) + h(start)  while open_set is not empty:      current ← node in open_set with lowest f(n)      if current = goal:         return reconstruct_path(came_from, current)      remove current from open_set      for each neighbor of current:          tentative_g ← g(current) + cost(current, neighbor)          if neighbor not visited OR tentative_g &lt; g(neighbor):              came_from[neighbor] ← current             g(neighbor) ← tentative_g             f(neighbor) ← g(neighbor) + h(neighbor)              if neighbor not in open_set:                 add neighbor to open_set  return failure</pre> |   |   |   |   |  |   |     |
|     | (OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |   |   |   |  |   |     |
| (b) | <p>Perform A* search to find the solution path for the following 8-puzzle problem</p> <table border="1"><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>8</td><td></td><td>4</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 1 | 2 | 3 | 8 |  | 4 | CO2 |
| 1   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 3 |   |   |   |  |   |     |
| 8   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4 |   |   |   |  |   |     |

|   |   |   |
|---|---|---|
| 2 | 8 | 3 |
| 1 | 6 | 4 |
| 7 |   | 5 |

|   |   |   |
|---|---|---|
| 7 | 6 | 5 |
|---|---|---|

INITIAL STATE

FINAL STATE



REACHED

THE FINAL STATE AT  $G=5, H=0, F=5$ .

13. (a) i) Describe the concept of Alpha-Beta pruning and its role in optimizing Minimax search. (5 Marks)
- ii) Construct a three-level game tree and demonstrate the pruning process in Alpha-Beta search. (8 Marks)
- i) Alpha-Beta Pruning is an optimization technique that significantly improves the efficiency of the Minimax algorithm in two-player decision-making games by reducing the number of nodes evaluated in the search tree.
- What is Alpha-Beta Pruning?**
- Alpha-beta pruning eliminates branches in the game tree that do not influence the final decision, thus saving computation time.

CO3



- Alpha ( $\alpha$ ): The best (highest) value that the maximizing player can guarantee so far.
- Beta ( $\beta$ ): The best (lowest) value that the minimizing player can guarantee so far.

When evaluating nodes, if it is found that a particular branch cannot possibly affect the outcome (i.e., the maximizer or minimizer already has a better alternative), the algorithm "prunes" or disregards that branch—skipping unnecessary work.

### How Alpha-Beta Pruning Works

- Initialization: Alpha is set to negative infinity, beta to positive infinity at the root.
- Node Evaluation:
  - For each child, recursively evaluate using Minimax, updating alpha and beta as you go.
  - If at any point,  $\alpha \geq \beta$  (for maximizer nodes) or  $\beta \leq \alpha$  (for minimizer nodes), prune further children—these cannot improve the final outcome.

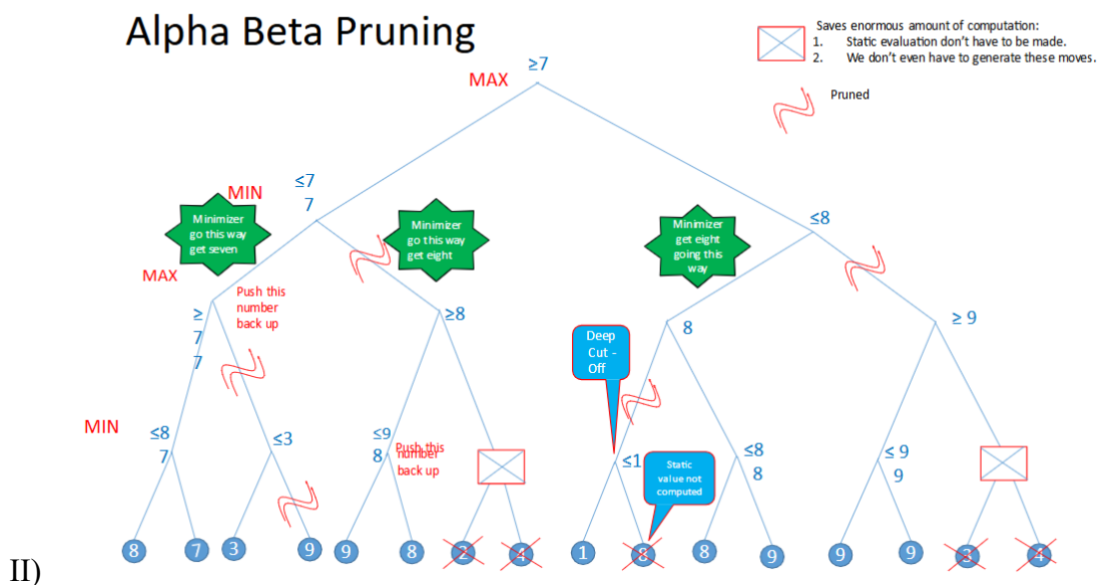
### Example

Suppose a game tree with several choices.

- If while exploring, the maximizing player finds a move with value 10, and another move with value less than 10 is encountered, there's no need to fully explore the second move's subtree if the minimizer can already assure a loss with lower value.

### How It Optimizes Minimax

- Pruning Irrelevant Branches: Only branches that can possibly affect the final choice are explored. Irrelevant moves for both player and opponent are skipped.
- Same Optimality, Less Work: The algorithm still guarantees the same optimal move as the basic minimax, but with far fewer evaluations—enabling deeper search within the same computation budget.
- Efficiency: In the best case (when moves are optimally ordered), alpha-beta pruning can reduce the time complexity from  $O(bd)$  to  $O(bd/2)$ , where  $b$  is the branching factor and  $d$  the tree depth



(OR)



Does not model real-world uncertainty well Models real-world uncertainty effectively

## 6. Expectiminimax Algorithm

In deterministic games → **Minimax**

In stochastic games → **Expectiminimax**

Three types of nodes:

- **Max nodes** (AI player)
- **Min nodes** (Opponent)
- **Chance nodes** (Probability)

Expected value is calculated as:

Expected Value =  $\sum P(s') \times U(s')$

The best move is the one with **maximum expected utility**.

## 7. Relationship with Decision-Making Under Uncertainty

Stochastic games are directly connected to **decision theory in AI** because:

### 1. Uncertain Outcomes

Actions do not guarantee fixed results. Agents must consider probabilities.

### 2. Expected Utility Principle

Rational agents choose actions that maximize expected utility.

### 3. Multi-Agent Environment

Unlike single-agent uncertainty models, stochastic games consider:

- Competition
- Strategic interaction

### 4. Connection to MDP

- **Markov Decision Process (MDP)** → Single-agent stochastic problem
- **Stochastic Game** → Multi-agent extension of MDP

### 5. Real-World Applications

- Autonomous vehicles
- Financial trading
- Security systems
- Robotics

## 8. Importance in AI

Stochastic games help AI systems:

- Make optimal decisions in uncertain environments
- Balance risk and reward
- Model competition and cooperation
- Handle partially observable scenarios

They combine:

- Game Theory
- Probability Theory
- Decision Theory

ii) Describe value iteration and policy iteration algorithms used in Markov Decision Processes with an example. (7 Marks)

A **Markov Decision Process (MDP)** is a mathematical model used for **sequential decision-making under uncertainty**.

An MDP consists of:

- **States (S)**
- **Actions (A)**
- **Transition Probability  $P(s' | s, a)$**
- **Reward Function  $R(s, a)$**
- **Discount Factor ( $\gamma$ )**

Objective:

To find an **optimal policy ( $\pi^*$ )** that maximizes expected cumulative reward.

**Value Iteration Algorithm**

**Definition:**

|                                  | <p>Value Iteration is a dynamic programming method that <b>computes optimal state values directly</b> and then derives the optimal policy.</p> <p><b>Bellman Optimality Equation:</b></p> $V(s)=\max_a \sum_{s'} P(s' s,a)[R(s,a)+\gamma V(s')]$ <p>Initialize <math>V(s)=0</math> for all states.</p> <ol style="list-style-type: none"><li>1. Update values using Bellman equation.</li><li>2. Repeat until values converge.</li><li>3. Extract policy using:</li></ol> $\pi(s)=\arg\max_a \sum_{s'} P(s' s,a)[R+\gamma V(s')]$ <p>It repeatedly improves value estimates until they stabilize.</p> <p><b>Policy Iteration Algorithm</b></p> <p><b>Definition:</b></p> <p>Policy Iteration alternates between:</p> <ul style="list-style-type: none"><li>• <b>Policy Evaluation</b></li><li>• <b>Policy Improvement</b></li></ul> <p><b>Steps:</b></p> <p><b>Step 1: Initialize</b></p> <p>Choose any policy <math>\pi</math>.</p> <p><b>Step 2: Policy Evaluation</b></p> <p>Compute value function <math>V_{\pi}(s)</math></p> $V_{\pi}(s)=\sum_{s'} P(s' s,\pi(s))[R+\gamma V_{\pi}(s')]$ <p><b>Step 3: Policy Improvement</b></p> $\pi'(s)=\arg\max_a \sum_{s'} P(s' s,a)[R+\gamma V_{\pi}(s')]$ <p><b>Step 4:</b></p> <p>Repeat until policy no longer changes.</p> <p>It improves policy step-by-step until optimal policy is found.</p> <p><b>4. Example (Simple Grid World)</b></p> <p>Consider a small grid:</p> <ul style="list-style-type: none"><li>• States: Cells in grid</li><li>• Actions: Up, Down, Left, Right</li><li>• Reward: +10 at goal state</li><li>• Movement has probability 0.8 correct, 0.2 random</li></ul> <p><b>Value Iteration:</b></p> <p>Updates each state value until optimal path to goal is found.</p> <p><b>Policy Iteration:</b></p> <p>Start with random movement policy → evaluate → improve direction toward goal → repeat → final optimal path.</p> <p><b>Difference Between Value Iteration and Policy Iteration</b></p> <table><thead><tr><th>Value Iteration</th><th>Policy Iteration</th></tr></thead><tbody><tr><td>Updates values directly</td><td>Improves policy directly</td></tr><tr><td>One step process</td><td>Two-step process</td></tr><tr><td>Slower convergence in some cases</td><td>Often converges faster</td></tr><tr><td>Less memory</td><td>Slightly more computation per iteration</td></tr></tbody></table> <p>Both algorithms are dynamic programming methods used to compute <b>optimal policies in MDPs</b>.</p> <ul style="list-style-type: none"><li>• <b>Value Iteration</b> focuses on value updates.</li><li>• <b>Policy Iteration</b> alternates between evaluation and improvement.</li></ul> <p>They are fundamental for decision-making under uncertainty in AI systems such as robotics, planning, and reinforcement learning.</p> | Value Iteration | Policy Iteration | Updates values directly | Improves policy directly | One step process | Two-step process | Slower convergence in some cases | Often converges faster | Less memory | Slightly more computation per iteration |  |
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| Value Iteration                  | Policy Iteration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                  |                         |                          |                  |                  |                                  |                        |             |                                         |  |
| Updates values directly          | Improves policy directly                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                 |                  |                         |                          |                  |                  |                                  |                        |             |                                         |  |
| One step process                 | Two-step process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                 |                  |                         |                          |                  |                  |                                  |                        |             |                                         |  |
| Slower convergence in some cases | Often converges faster                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 |                  |                         |                          |                  |                  |                                  |                        |             |                                         |  |
| Less memory                      | Slightly more computation per iteration                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                 |                  |                         |                          |                  |                  |                                  |                        |             |                                         |  |
|                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                 |                  |                         |                          |                  |                  |                                  |                        |             |                                         |  |

| 14.                 | (a)                      | <p>i) Compare Propositional Logic and First-Order Logic in terms of expressiveness and representation. (6 Marks)</p> <table><tr><th>Basis</th><th>Propositional Logic (PL)</th><th>First-Order Logic (FOL)</th></tr><tr><td>Basic Unit</td><td>Propositions (P, Q, R)</td><td>Predicates with objects</td></tr><tr><td>Expressiveness</td><td>Limited</td><td>Highly expressive</td></tr><tr><td>Variables</td><td>Not allowed</td><td>Allowed</td></tr><tr><td>Quantifiers</td><td>Not available</td><td><math>\forall</math> (for all), <math>\exists</math> (there exists)</td></tr><tr><td>Representation</td><td>Whole facts only</td><td>Facts about objects &amp; relations</td></tr><tr><td>Structure</td><td>No internal structure</td><td>Has internal structure</td></tr><tr><td>Example</td><td>P: "It is raining"</td><td>Rain(x), Student(x)</td></tr><tr><td>Real-world modeling</td><td>Simple domains</td><td>Complex domains with relationships</td></tr><tr><td>Knowledge reuse</td><td>Low</td><td>High (uses variables &amp; general rules)</td></tr></table> <p><b>Expressiveness</b></p> <ul style="list-style-type: none"><li>• <b>Propositional Logic</b> can represent only simple true/false statements.<br/>Example:<br/><math>P \rightarrow Q</math></li><li>• <b>First-Order Logic</b> can represent:<ul style="list-style-type: none"><li>○ Objects</li><li>○ Properties</li><li>○ Relationships</li><li>○ General rules</li></ul></li></ul> <p>Example:<br/><math>\forall x (\text{Student}(x) \rightarrow \text{Studies}(x))</math><br/>Thus, <b>FOL is more expressive than PL.</b><br/>Propositional Logic is suitable for <b>simple, finite domains</b>, whereas First-Order Logic is suitable for <b>complex knowledge representation systems</b> involving objects and relationships.</p> <p>ii) Explain the syntax and semantics of First-Order Logic with examples and describe how it is used in knowledge representation systems. (7 Marks)</p> <p>First-Order Logic extends propositional logic by adding:</p> <ul style="list-style-type: none"><li>• Variables</li><li>• Quantifiers</li><li>• Predicates</li><li>• Functions</li></ul> <p>It allows representation of <b>objects, properties, relations, and rules.</b></p> <p><b>2. Syntax of First-Order Logic</b><br/>Syntax defines the <b>structure of valid sentences.</b><br/><b>Basic Components:</b></p> <ol style="list-style-type: none"><li>1. <b>Constants</b> – Represent specific objects<br/>Example: John, India</li><li>2. <b>Variables</b> – x, y, z</li><li>3. <b>Predicates</b> – Represent properties or relations<br/>Example:<br/>Student(x)<br/>Loves(x, y)</li><li>4. <b>Functions</b> – Map objects to objects<br/>Example: Father(x)</li><li>5. <b>Logical Connectives</b> – <math>\wedge, \vee, \neg, \rightarrow</math></li><li>6. <b>Quantifiers</b><ul style="list-style-type: none"><li>○ Universal: <math>\forall x</math> (for all x)</li><li>○ Existential: <math>\exists x</math> (there exists x)</li></ul></li></ol> <p><b>Example of Syntax:</b></p> | Basis | Propositional Logic (PL) | First-Order Logic (FOL) | Basic Unit | Propositions (P, Q, R) | Predicates with objects | Expressiveness | Limited | Highly expressive | Variables | Not allowed | Allowed | Quantifiers | Not available | $\forall$ (for all), $\exists$ (there exists) | Representation | Whole facts only | Facts about objects & relations | Structure | No internal structure | Has internal structure | Example | P: "It is raining" | Rain(x), Student(x) | Real-world modeling | Simple domains | Complex domains with relationships | Knowledge reuse | Low | High (uses variables & general rules) | CO4 |
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| Basis               | Propositional Logic (PL) | First-Order Logic (FOL)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Basic Unit          | Propositions (P, Q, R)   | Predicates with objects                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Expressiveness      | Limited                  | Highly expressive                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Variables           | Not allowed              | Allowed                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Quantifiers         | Not available            | $\forall$ (for all), $\exists$ (there exists)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Representation      | Whole facts only         | Facts about objects & relations                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Structure           | No internal structure    | Has internal structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Example             | P: "It is raining"       | Rain(x), Student(x)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Real-world modeling | Simple domains           | Complex domains with relationships                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |
| Knowledge reuse     | Low                      | High (uses variables & general rules)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |       |                          |                         |            |                        |                         |                |         |                   |           |             |         |             |               |                                               |                |                  |                                 |           |                       |                        |         |                    |                     |                     |                |                                    |                 |     |                                       |     |

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|  |     | <ol style="list-style-type: none"> <li>1. <math>\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))</math></li> <li>2. <math>\text{Human}(\text{Socrates})</math></li> <li>3. <math>\text{Mortal}(\text{Socrates})</math></li> </ol> <p><b>3. Semantics of First-Order Logic</b><br/>Semantics defines the <b>meaning</b> of sentences.<br/>It includes:</p> <ol style="list-style-type: none"> <li>1. <b>Domain</b> – Set of objects being discussed</li> <li>2. <b>Interpretation</b> – Assigns meaning to: <ul style="list-style-type: none"> <li>○ Constants <math>\rightarrow</math> Objects</li> <li>○ Predicates <math>\rightarrow</math> Relations</li> <li>○ Functions <math>\rightarrow</math> Mappings</li> </ul> </li> </ol> <p><b>Example of Semantics:</b><br/>Statement:<br/><math>\forall x (\text{Student}(x) \rightarrow \text{Smart}(x))</math><br/>Meaning:<br/>For every object in the domain, if it is a student, then it is smart.<br/>Truth depends on:</p> <ul style="list-style-type: none"> <li>• The domain chosen</li> <li>• The interpretation assigned</li> </ul> <p><b>4. Use of FOL in Knowledge Representation Systems</b><br/>First-Order Logic is used in AI systems for:</p> <ol style="list-style-type: none"> <li>1. <b>Knowledge-Based Agents</b> <ul style="list-style-type: none"> <li>• Represent facts and rules</li> <li>• Perform logical inference</li> </ul> </li> <li>2. <b>Expert Systems</b> <ul style="list-style-type: none"> <li>• Encode domain knowledge</li> <li>• Derive conclusions using inference rules</li> </ul> </li> <li>3. <b>Database Systems</b> <ul style="list-style-type: none"> <li>• Represent relationships between entities</li> </ul> </li> <li>4. <b>Automated Theorem Proving</b> <ul style="list-style-type: none"> <li>• Forward chaining</li> <li>• Backward chaining</li> <li>• Resolution</li> </ul> </li> </ol> <p><b>5. Example in Knowledge Representation</b><br/>Knowledge Base:</p> <ol style="list-style-type: none"> <li>1. <math>\forall x (\text{Doctor}(x) \rightarrow \text{Educated}(x))</math></li> <li>2. <math>\text{Doctor}(\text{Ravi})</math></li> </ol> <p>Using inference:<br/><math>\text{Educated}(\text{Ravi})</math><br/>This shows how FOL supports <b>logical reasoning</b>.</p> <ul style="list-style-type: none"> <li>• Syntax defines the structure of FOL sentences.</li> <li>• Semantics defines their meaning.</li> <li>• FOL is powerful for representing structured knowledge.</li> <li>• It is widely used in AI for reasoning, expert systems, and intelligent agents.</li> </ul> |     |
|  |     | <b>(OR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |     |
|  | (b) | i) Compare and contrast Forward Chaining and Backward Chaining inference methods used in logical agents. (6 Marks)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | CO4 |

i) Forward Chaining and Backward Chaining are both inference methods used in AI for reasoning and decision-making, but they differ primarily in their approach and direction of reasoning.

Forward Chaining is a data-driven method that starts with known facts in the knowledge base and applies inference rules to derive new facts until it reaches a goal. It operates in a bottom-up manner, exploring all possible conclusions by repeatedly applying rules to the current facts. This approach suits environments where all available data is processed continuously to discover new knowledge. For example, in a medical diagnosis system, based on symptoms (facts), it can infer possible diseases by progressing forward through rules. Forward chaining uses a breadth-first search strategy and is efficient when multiple conclusions can be drawn simultaneously but can be slower and less focused if the goal is specific.

Backward Chaining, on the other hand, is a goal-driven method that begins with a specific goal or hypothesis and works backward to verify if known facts support it. It breaks down the goal into sub-goals recursively, searching for supporting facts or rules that confirm the hypothesis. This top-down approach is efficient for targeted problem-solving, like troubleshooting or legal reasoning, where the goal is clear, and backward chaining only explores relevant rules. For example, a doctor suspecting a disease starts from the diagnosis goal and looks backward for symptoms to confirm it. Backward chaining uses a depth-first search strategy and generally requires less memory but can be complex due to recursive reasoning.

| Feature           | Forward Chaining                             | Backward Chaining                   |
|-------------------|----------------------------------------------|-------------------------------------|
| Direction         | Bottom-up: from facts to conclusions         | Top-down: from goal to facts        |
| Approach          | Data-driven                                  | Goal-driven                         |
| Starting Point    | Known facts                                  | Specific goal/hypothesis            |
| Search Strategy   | Breadth-first search                         | Depth-first search                  |
| Use Case Examples | Real-time monitoring, recommendation systems | Diagnostic systems, theorem proving |
| Efficiency        | Can be slower due to exploring all facts     | More efficient for focused goals    |
| Memory Usage      | Consumes more memory                         | Less memory intensive               |

|            |                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |            |                      |                               |  |
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|            |                      | <table><tr><td>Complexity</td><td>Simpler to implement</td><td>More complex due to recursion</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Complexity | Simpler to implement | More complex due to recursion |  |
| Complexity | Simpler to implement | More complex due to recursion                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |            |                      |                               |  |
|            |                      | <p>ii) Construct a simple knowledge base and demonstrate the Forward Chaining inference procedure step-by-step to derive a valid conclusion. (7 Marks)</p> <p>Facts:</p> <ul style="list-style-type: none"><li>• Fritz croaks</li><li>• Fritz eats flies</li></ul> <p>Rules:</p> <ol style="list-style-type: none"><li>1. If X croaks and X eats flies, then X is a frog.</li><li>2. If X is a frog, then X is green.</li><li>3. If X chirps and X sings, then X is a canary.</li><li>4. If X is a canary, then X is blue.</li></ol> <p><b>Goal:</b><br/>Determine Fritz's color.</p> <p><b>Forward Chaining Process:</b></p> <ul style="list-style-type: none"><li>• Step 1: Start with known facts: "Fritz croaks" and "Fritz eats flies."</li><li>• Step 2: Check rules to see which can be applied:<ul style="list-style-type: none"><li>• Rule 1: "X croaks and X eats flies → X is a frog" applies since Fritz satisfies both conditions.</li></ul></li><li>• Step 3: Add the conclusion "Fritz is a frog" to the facts.</li><li>• Step 4: Now with "Fritz is a frog," check other rules:<ul style="list-style-type: none"><li>• Rule 2: "X is a frog → X is green" applies.</li></ul></li><li>• Step 5: Add "Fritz is green" to the facts.</li></ul> <p>No other rules apply given the current facts, so the process stops.</p> <p>Using forward chaining, Fritz can be logically concluded to be green.</p> <p>This simple example illustrates forward chaining as a data-driven, bottom-up inference technique that starts from facts, applies inference rules to derive new facts, and ultimately reaches a conclusion stepwise.</p> |            |                      |                               |  |
| 15.        | (a)                  | <p>a) What is Hierarchical Planning? Explain its advantages over classical planning. (6 Marks)</p> <p><b>Hierarchical Planning</b> (also called Hierarchical Task Network – HTN planning) is a planning approach in which a complex problem is <b>decomposed into smaller subtasks</b> arranged in a hierarchy.</p> <p>Instead of directly searching for primitive actions (as in classical planning), it:</p> <ul style="list-style-type: none"><li>• Starts with a high-level goal</li><li>• Breaks it into intermediate tasks</li><li>• Further decomposes into executable primitive actions</li></ul> <p>Thus, planning is done <b>top-down</b>.</p> <p>Goal: <i>Organize a conference</i></p> <p>Hierarchical breakdown:</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO5        |                      |                               |  |



1. Organize conference
  - Arrange venue
  - Invite speakers
  - Prepare schedule

Each task can further be broken into smaller actions.

### 3. Classical Planning

In classical planning:

- Environment is deterministic
- Fully observable
- Actions have fixed effects
- Planning searches directly in state space

#### Advantages of Hierarchical Planning over Classical Planning

| Hierarchical Planning                  | Classical Planning                |
|----------------------------------------|-----------------------------------|
| Decomposes complex tasks               | Direct state-space search         |
| Reduces search space                   | Large search space                |
| More structured                        | Less structured                   |
| Uses domain knowledge                  | Less domain guidance              |
| More efficient for real-world problems | Efficient only for small problems |

#### Key Advantages:

1. **Reduces Complexity** – By dividing large problems into manageable subtasks.
2. **Improves Efficiency** – Smaller search space.
3. **Incorporates Domain Knowledge** – Uses predefined task decomposition.
4. **Closer to Human Planning** – Mimics human problem-solving style.

Hierarchical planning is more suitable for **complex real-world AI systems**, whereas classical planning is appropriate for simple deterministic environments.

b) Discuss planning in non-deterministic domains and explain how time, scheduling, and resource constraints are handled in AI planning systems. (7 Marks)

#### Planning in Non-Deterministic Domains

In non-deterministic domains:

- Actions may have **multiple possible outcomes**
- Results are uncertain
- Environment may change unexpectedly

Example:

A robot tries to pick an object → may succeed or fail.

### 2. Types of Planning in Non-Deterministic Domains

1. **Conditional Planning**
  - Plan includes branches for different outcomes.
  - Uses “if-then” structure.
2. **Contingency Planning**
  - Prepares for possible failures.
3. **Policy-Based Planning**
  - Creates mapping from states to actions (similar to MDP).

### 3. Handling Time in AI Planning

Time is handled using:

#### 1. Temporal Planning

- Actions have duration.
- Start and end times are considered.
- Allows overlapping actions.

Example:

Cooking tasks happening simultaneously.

|  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |     |
|--|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|  |     | <p><b>4. Scheduling in AI Planning</b></p> <p>Scheduling decides:</p> <ul style="list-style-type: none"> <li>• Order of tasks</li> <li>• Start time</li> <li>• Resource allocation</li> </ul> <p>It ensures:</p> <ul style="list-style-type: none"> <li>• No task conflicts</li> <li>• Deadlines are met</li> </ul> <p>Common techniques:</p> <ul style="list-style-type: none"> <li>• Constraint satisfaction</li> <li>• Heuristic search</li> </ul> <p><b>5. Handling Resource Constraints</b></p> <p>AI systems must consider limited resources such as:</p> <ul style="list-style-type: none"> <li>• Fuel</li> <li>• Money</li> <li>• Workers</li> <li>• Memory</li> </ul> <p>Methods used:</p> <ol style="list-style-type: none"> <li>1. <b>Constraint-Based Planning</b> <ul style="list-style-type: none"> <li>◦ Ensures resources are not overused.</li> </ul> </li> <li>2. <b>Linear Programming</b></li> <li>3. <b>Heuristic resource allocation</b></li> </ol> <p><b>6. Example</b></p> <p>In factory planning:</p> <ul style="list-style-type: none"> <li>• Machines (resources) are limited</li> <li>• Tasks have deadlines (time constraint)</li> <li>• Some tasks may fail (non-determinism)</li> </ul> <p>The planner creates:</p> <ul style="list-style-type: none"> <li>• Conditional plans</li> <li>• Time-based schedules</li> <li>• Resource-aware execution plans</li> </ul> <p>Planning in non-deterministic domains requires flexible strategies such as conditional and policy-based planning. AI planning systems handle <b>time, scheduling, and resource constraints</b> using temporal reasoning, constraint satisfaction, and optimization techniques to produce feasible and efficient plans.</p> |     |
|  |     | (OR)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |     |
|  | (b) | <p>a) Define Classical Planning and explain the basic structure of a planning problem. (6 Marks)</p> <p><b>Classical Planning</b> is a planning approach in AI where:</p> <ul style="list-style-type: none"> <li>• The environment is <b>fully observable</b></li> <li>• The environment is <b>deterministic</b></li> <li>• The environment is <b>static</b></li> <li>• Actions have <b>known and fixed effects</b></li> <li>• The agent has complete knowledge of the world</li> </ul> <p>The goal is to find a <b>sequence of actions</b> that transforms the initial state into a goal state.</p> <p><b>2. Basic Structure of a Classical Planning Problem</b></p> <p>A classical planning problem is defined as a tuple:<br/> <math>P=(S,A,I,G)</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | CO5 |

Where:

1. **S (States)**  
Set of all possible states of the world.
2. **A (Actions)**  
Set of available actions.  
Each action contains:
  - Preconditions
  - Effects (Add list and Delete list)
3. **I (Initial State)**  
The starting state of the system.
4. **G (Goal State)**  
Desired condition to be achieved.

### 3. Example

Example: Robot moving in a room

- Initial State: Robot at Room A
- Goal State: Robot at Room B
- Action: Move(A  $\rightarrow$  B)
  - Precondition: At(A)
  - Effect: At(B),  $\neg$ At(A)

The planner finds the sequence of actions to reach the goal.

Classical planning searches for a valid sequence of actions under deterministic and fully observable conditions using logical representation of states and actions.

b) Describe the main algorithms used in classical planning and discuss how heuristics improve planning efficiency. (7 Marks)

### Main Algorithms Used in Classical Planning

#### 1. State-Space Search

Planning is treated as a search problem.

Two approaches:

##### a) Forward State-Space Search (Progression)

- Start from initial state
- Apply actions until goal is reached

##### b) Backward State-Space Search (Regression)

- Start from goal
- Work backward to initial state

#### 2. Planning Graph (GraphPlan)

- Builds a layered graph of states and actions
- Alternates between proposition layers and action layers
- Efficiently detects goal reachability

#### 3. SAT-Based Planning

- Converts planning problem into Boolean satisfiability problem
- Uses SAT solvers to find action sequence

### 2. Role of Heuristics in Planning

Without heuristics:

- Search space becomes very large
- Planning becomes inefficient

Heuristics help by:

1. **Estimating distance to goal**
2. **Reducing search space**
3. **Guiding search toward promising states**

|                                | <div><h3>Examples of Heuristics</h3><div><div>1. Relaxed Problem Heuristic<ul style="list-style-type: none"><li>Ignore delete effects</li><li>Gives optimistic estimate</li></ul></div><div>2. Planning Graph Heuristic<ul style="list-style-type: none"><li>Uses level cost of goal</li></ul></div><div>3. A* Search Heuristic<ul style="list-style-type: none"><li><math>f(n) = g(n) + h(n)</math></li></ul></div></div><div><h3>3. How Heuristics Improve Efficiency</h3><table><thead><tr><th>Without Heuristic</th><th>With Heuristic</th></tr></thead><tbody><tr><td>Blind search</td><td>Guided search</td></tr><tr><td>Explores many states</td><td>Explores relevant states</td></tr><tr><td>High time complexity</td><td>Reduced computation time</td></tr><tr><td>Inefficient for large problems</td><td>Scales better</td></tr></tbody></table><p>Classical planning uses search-based and graph-based algorithms to find action sequences. Heuristics significantly improve planning efficiency by reducing the search space and guiding the planner toward optimal solutions.</p></div></div> <td></td> | Without Heuristic | With Heuristic | Blind search | Guided search | Explores many states | Explores relevant states | High time complexity | Reduced computation time | Inefficient for large problems | Scales better |  |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|----------------|--------------|---------------|----------------------|--------------------------|----------------------|--------------------------|--------------------------------|---------------|--|
| Without Heuristic              | With Heuristic                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                   |                |              |               |                      |                          |                      |                          |                                |               |  |
| Blind search                   | Guided search                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                   |                |              |               |                      |                          |                      |                          |                                |               |  |
| Explores many states           | Explores relevant states                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |                |              |               |                      |                          |                      |                          |                                |               |  |
| High time complexity           | Reduced computation time                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |                |              |               |                      |                          |                      |                          |                                |               |  |
| Inefficient for large problems | Scales better                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                   |                |              |               |                      |                          |                      |                          |                                |               |  |

### PART C

(1 x 15 = 15 marks)

(Case study/Comprehensive type Questions)

|    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |
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|    |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>CO</b> |
| 16 | (a) | <p>Consider the following sentences:</p> <ol style="list-style-type: none"> <li>All students who study hard pass the exam.</li> <li>Mathematics is a subject.</li> <li>Anyone who studies a subject and practices regularly succeeds.</li> <li>Alice studies Mathematics and practices daily.</li> <li>Hospitals provide medical treatment.</li> <li>Employees receive salaries.</li> <li>Salaries help people sustain their livelihood.</li> <li>Cats are animals.</li> <li>Anyone who owns a pet takes care of it.</li> <li>Electricity provides energy</li> </ol> <p>a) Represent the above knowledge using First-Order Predicate Logic. Clearly define the predicates, constants, and quantifiers used. (8 Marks)</p> <p>b) Convert all the predicate logic statements into clause form (CNF). Show all intermediate steps including:</p> <ul style="list-style-type: none"> <li>Elimination of implications</li> <li>Standardization of variables</li> <li>Skolemization</li> <li>Conversion to Conjunctive Normal Form</li> <li>Extraction of clauses</li> </ul> <p>(7 Marks)</p> <p><b>Representation using First-Order Predicate Logic (8 Marks)</b></p> <p><b>1. Predicates Used</b></p> <ul style="list-style-type: none"> <li>Student(x) – x is a student</li> <li>StudiesHard(x) – x studies hard</li> <li>Pass(x) – x passes exam</li> <li>Subject(x) – x is a subject</li> <li>Studies(x, y) – x studies subject y</li> </ul> | CO<br>4   |

|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |  |
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|  | <ul style="list-style-type: none"> <li>• PracticesRegularly(x) – x practices regularly</li> <li>• Succeeds(x) – x succeeds</li> <li>• Hospital(x) – x is a hospital</li> <li>• Provides(x, y) – x provides y</li> <li>• Employee(x) – x is an employee</li> <li>• Receives(x, y) – x receives y</li> <li>• Helps(x, y) – x helps y</li> <li>• SustainLivelihood(x) – x sustains livelihood</li> <li>• Cat(x) – x is a cat</li> <li>• Animal(x) – x is an animal</li> <li>• Owns(x, y) – x owns y</li> <li>• Pet(y) – y is a pet</li> <li>• TakesCare(x, y) – x takes care of y</li> <li>• Electricity(x) – x is electricity</li> <li>• Energy(y) – y is energy</li> </ul> <p><b>2. Constants Used</b></p> <ul style="list-style-type: none"> <li>• Alice</li> <li>• Mathematics</li> <li>• MedicalTreatment</li> <li>• Salary</li> <li>• Livelihood</li> <li>• Energy</li> </ul> <p><b>3. FOL Representation</b></p> <p>i) <math>\forall x [(Student(x) \wedge StudiesHard(x)) \rightarrow Pass(x)]</math><br/> ii) <math>Subject(Mathematics)</math><br/> iii) <math>\forall x \forall y [(Studies(x,y) \wedge Subject(y) \wedge PracticesRegularly(x)) \rightarrow Succeeds(x)]</math><br/> iv) <math>Studies(Alice, Mathematics) \wedge PracticesRegularly(Alice)</math><br/> v) <math>\forall x [Hospital(x) \rightarrow Provides(x, MedicalTreatment)]</math><br/> vi) <math>\forall x [Employee(x) \rightarrow Receives(x, Salary)]</math><br/> vii) <math>\forall x [Receives(x, Salary) \rightarrow SustainLivelihood(x)]</math><br/> viii) <math>\forall x [Cat(x) \rightarrow Animal(x)]</math><br/> ix) <math>\forall x \forall y [(Owns(x,y) \wedge Pet(y)) \rightarrow TakesCare(x,y)]</math><br/> x) <math>\forall x [Electricity(x) \rightarrow Provides(x, Energy)]</math></p> <p><b>b) Conversion to Clause Form (CNF) (7 Marks)</b><br/> We convert only statements containing implications (facts remain same).</p> |  |
|  | <p><b>Step 1: Eliminate Implications (<math>\rightarrow</math>)</b><br/> Rule:<br/> <math>A \rightarrow B</math> becomes <math>\neg A \vee B</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
|  | <p><b>(i)</b><br/> <math>\forall x [\neg(Student(x) \wedge StudiesHard(x)) \vee Pass(x)]</math><br/> Apply De Morgan:<br/> <math>\forall x [\neg Student(x) \vee \neg StudiesHard(x) \vee Pass(x)]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |  |
|  | <p><b>(iii)</b><br/> <math>\forall x \forall y [\neg(Studies(x,y) \wedge Subject(y) \wedge PracticesRegularly(x)) \vee Succeeds(x)]</math><br/> Apply De Morgan:<br/> <math>\forall x \forall y [\neg Studies(x,y) \vee \neg Subject(y) \vee \neg PracticesRegularly(x) \vee Succeeds(x)]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |  |
|  | <p><b>(v)</b><br/> <math>\forall x [\neg Hospital(x) \vee Provides(x, MedicalTreatment)]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |  |

|  |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
|--|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------|
|  |     | <p><b>(vi)</b><br/> <math>\forall x [\neg \text{Employee}(x) \vee \text{Receives}(x, \text{Salary})]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |         |
|  |     | <p><b>(vii)</b><br/> <math>\forall x [\neg \text{Receives}(x, \text{Salary}) \vee \text{SustainLivelihood}(x)]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |         |
|  |     | <p><b>(viii)</b><br/> <math>\forall x [\neg \text{Cat}(x) \vee \text{Animal}(x)]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
|  |     | <p><b>(ix)</b><br/> <math>\forall x \forall y [\neg \text{Owns}(x, y) \vee \neg \text{Pet}(y) \vee \text{TakesCare}(x, y)]</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |         |
|  |     | <p><b>(x)</b><br/> <math>\forall x [\neg \text{Electricity}(x) \vee \text{Provides}(x, \text{Energy})]</math></p> <p><b>Step 2: Standardization of Variables</b><br/> All quantified variables are already uniquely named (x, y).<br/> No change required.</p> <p><b>Step 3: Skolemization</b><br/> There are <b>no existential quantifiers</b>, so no Skolem functions needed.</p> <p><b>Step 4: Drop Universal Quantifiers</b><br/> In CNF, universal quantifiers are dropped.</p> <p><b>Step 5: Extract Clauses</b><br/> Final Clause Set:</p> <ol style="list-style-type: none"> <li>1. <math>\neg \text{Student}(x) \vee \neg \text{StudiesHard}(x) \vee \text{Pass}(x)</math></li> <li>2. <math>\text{Subject}(\text{Mathematics})</math></li> <li>3. <math>\neg \text{Studies}(x, y) \vee \neg \text{Subject}(y) \vee \neg \text{PracticesRegularly}(x) \vee \text{Succeeds}(x)</math></li> <li>4. <math>\text{Studies}(\text{Alice}, \text{Mathematics})</math></li> <li>5. <math>\text{PracticesRegularly}(\text{Alice})</math></li> <li>6. <math>\neg \text{Hospital}(x) \vee \text{Provides}(x, \text{MedicalTreatment})</math></li> <li>7. <math>\neg \text{Employee}(x) \vee \text{Receives}(x, \text{Salary})</math></li> <li>8. <math>\neg \text{Receives}(x, \text{Salary}) \vee \text{SustainLivelihood}(x)</math></li> <li>9. <math>\neg \text{Cat}(x) \vee \text{Animal}(x)</math></li> <li>10. <math>\neg \text{Owns}(x, y) \vee \neg \text{Pet}(y) \vee \text{TakesCare}(x, y)</math></li> <li>11. <math>\neg \text{Electricity}(x) \vee \text{Provides}(x, \text{Energy})</math></li> </ol> |         |
|  |     | <b>(OR)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |         |
|  | (b) | <p>i) Explain the resolution principle in first-order logic and demonstrate its application with an example. (8 Marks)</p> <p>The resolution principle in first-order logic is a fundamental inference rule used in automated theorem proving to derive conclusions or prove statements by refutation (proof by contradiction). It works by resolving pairs of clauses that contain complementary literals (a literal and its negation), applying unification to make the literals identical if necessary, and producing a new clause called a <i>resolvent</i>. Repeating this process systematically can lead to deriving an empty clause, which represents a contradiction, thereby proving the original assertion true.</p> <p><b>Resolution Principle in First-Order Logic</b></p> <ul style="list-style-type: none"> <li>• First, all formulas are converted into clausal form (conjunctive normal form without quantifiers, after Skolemization and standardization).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO<br>4 |

- The goal to be proved is negated and added to the knowledge base.
- The algorithm repeatedly applies the resolution rule: find two clauses with complementary literals, unify variables if needed, and combine the rest of the literals into a new clause called the *resolvent*.
- This process continues until either (a) an empty clause is generated (indicating a contradiction, so the original goal is proved) or (b) no new resolvents can be generated (the goal is not provable).

### Example Demonstration

Problem:

Given the knowledge base:

1. All humans are mortal:  $\forall x(\text{Human}(x) \rightarrow \text{Mortal}(x))$
2. Socrates is a human:  $\text{Human}(\text{Socrates})$

Prove: Socrates is mortal:  $\text{Mortal}(\text{Socrates})$

Step 1: Convert to clauses

- $\neg \text{Human}(x) \vee \text{Mortal}(x)$
- $\text{Human}(\text{Socrates})$
- Negation of goal (for refutation):  $\neg \text{Mortal}(\text{Socrates})$

Step 2: Apply resolution

- Resolve  $\neg \text{Human}(x) \vee \text{Mortal}(x)$  and  $\text{Human}(\text{Socrates})$  by unifying  $x = \text{Socrates}$   
Resolvent:  $\text{Mortal}(\text{Socrates})$
- Resolve  $\text{Mortal}(\text{Socrates})$  and  $\neg \text{Mortal}(\text{Socrates})$   
Resolvent: empty clause (contradiction found)

Step 3: Conclusion

The empty clause indicates that the negation of the goal leads to contradiction, hence  $\text{Mortal}(\text{Socrates})$  is true.

The resolution principle effectively automates logical inference by systematically combining clauses to derive contradictions and prove statements. Its key components include:

- Clauses and Literals: Basic units of resolution.
- Unification: Finding substitutions to make literals identical.
- Resolution rule: Combining clauses to infer new clauses.
- Refutation completeness: Guaranteed to find contradictions if the knowledge base entails the goal.

This proof technique is foundational to logic programming, automated reasoning, and many AI applications.

ii) Compare resolution with forward and backward chaining. Discuss how resolution helps in automated theorem proving. (6 Marks)

| Aspect         | Resolution                               | Forward Chaining       | Backward Chaining     |
|----------------|------------------------------------------|------------------------|-----------------------|
| Reasoning Type | Refutation-based, proof by contradiction | Data-driven, bottom-up | Goal-driven, top-down |

|                  |                                                                   |                                                     |                                                           |
|------------------|-------------------------------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------|
| Input            | Clausal form of knowledge + negated goal                          | Known facts and rules                               | Goal/hypothesis + rules                                   |
| Process          | Resolves pairs of clauses to derive contradictions or prove goals | Starts from facts, applies rules to infer new facts | Starts from goal, works backward to find supporting facts |
| Search Strategy  | Systematic resolution of clauses                                  | Breadth-first search of rule applications           | Depth-first search of subgoals                            |
| Output           | Empty clause (contradiction) or proof of goal                     | New inferred facts or goal conclusions              | Verification if goal can be derived                       |
| Automation & Use | Widely used in automated theorem proving and proofs               | Used in expert systems, real-time decision-making   | Used in diagnostics, troubleshooting, hypothesis testing  |

### How Resolution Helps in Automated Theorem Proving

- Refutation completeness: Resolution is sound and complete for first-order logic, meaning if the goal logically follows, resolution will find a proof by deriving a contradiction.
- Uniformity: It uses a single, uniform inference rule (resolution), simplifying implementation of theorem provers compared to chaining that relies on rule-based reasoning engines.
- Unification: Resolution incorporates unification to handle variables systematically, allowing general and powerful reasoning over first-order logic statements.
- Systematic search: Automated theorem provers use resolution to exhaustively attempt clause resolutions, ensuring all logical possibilities are explored.
- Proof generation: By tracking resolution steps, theorem provers can generate explicit proofs, useful for verification and explanation.

In contrast, forward and backward chaining are more specialized inference strategies suited for rule-based expert systems and decision-making, focusing on chaining rules rather than exhaustive logical proofs.

Resolution is fundamental for automated logical proofs and formal verification in AI, providing a complete, systematic approach for theorem proving, while forward and backward chaining are practical inference methods optimized for knowledge-based reasoning and goal-directed problem-solving. Resolution, forward chaining, and



|  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
|  | <p>backward chaining are all inference methods used in AI but differ fundamentally in their approach and application:</p> <ul style="list-style-type: none"> <li>Resolution is a refutation-based, logic-driven inference method that works by deriving contradictions through systematic application of the resolution rule on clauses. It is complete and sound for first-order logic and forms the basis for automated theorem proving by proving statements via contradiction. Resolution handles variables through unification and performs an exhaustive search to show that a negation of the goal leads to contradiction.</li> <li>Forward Chaining is a data-driven inference technique starting from known facts and applying inference rules to derive new facts until a conclusion is reached. It uses a breadth-first search strategy and is suited for domains with abundant data where multiple conclusions need to be drawn. It is used mainly in expert systems and monitoring environments.</li> <li>Backward Chaining is a goal-driven inference method that starts with specific goals or hypotheses and works backward to find facts that support the goals. It employs a depth-first search approach and is efficient for targeted problem-solving like diagnostics or troubleshooting.</li> </ul> <p>Resolution helps in automated theorem proving by:</p> <ul style="list-style-type: none"> <li>Offering a uniform inference rule that can be algorithmically applied to any clause in first-order logic.</li> <li>Utilizing unification to systematically deal with variables and substitutions.</li> <li>Ensuring refutation completeness, meaning if a statement is logically entailed, resolution will find a proof.</li> <li>Enabling automated systems to generate explicit proofs or contradictions, crucial for verification and verification tasks.</li> </ul> <p>In contrast, forward and backward chaining are more heuristic, rule-based reasoning engines optimized for practical problem-solving rather than exhaustive logical proof generation.</p> <p>Hence, resolution forms the theoretical and algorithmic foundation for automated reasoning, while forward and backward chaining are application-oriented techniques for expert systems and decision-making.</p> |  |
|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|

|                  | For Set 3 QP |       |       |       |       |       |       |       |       |       |          |       |
|------------------|--------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|----------|-------|
|                  | Part - A     |       |       |       |       |       |       |       |       |       |          |       |
| Question No.     | 1            | 2     | 3     | 4     | 5     | 6     | 7     | 8     | 9     | 10    |          |       |
| Knowledge Level  | K1           | K1    | K1    | K1    | K1    | K1    | K2    | K1    | K1    | K1    |          |       |
| Difficulty Level | 1            | 1     | 1     | 1     | 1     | 1     | 2     | 1     | 1     | 1     |          |       |
|                  | Part - B     |       |       |       |       |       |       |       |       |       | Part - C |       |
| Question No.     | 11(a)        | 11(b) | 12(a) | 12(b) | 13(a) | 13(b) | 14(a) | 14(b) | 15(a) | 15(b) | 16(a)    | 16(b) |
| Knowledge Level  | K4           | K3    | K3    | K3    | K4    | K4    | K4    | K4    | K4    | K4    | K4       | K4    |
| Difficulty Level | 4            | 4     | 4     | 4     | 3     | 4     | 4     | 3     | 4     | 4     | 4        | 4     |